Machine Dialectology: Heterogeneous Language-Model Societies Develop, Teach and Route Reasoning Dialects

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Abstract

Large language models are increasingly deployed as heterogeneous systems of solvers, routers, critics, and tool users; yet, their intermediate reasoning logic is still expressed as human-facing prose that may not be optimal from computational perspective. We introduce Machine Dialectology (MDia), a test-time framework for studying and exploiting reusable machine dialects, *i.e.*, compact Language Symbolism Frameworks (LSFs) generated by speaker models and interpreted by listener models. MDia records each speaker–listener–dialect–task event in a dynamic archive, estimates social profiles such as publicness, openness, resistance, and teaching advantage, and uses these profiles to route dialects under accuracy and token budgets. Across eight benchmark columns spanning strong reasoning, function calling, code-output prediction, multi-hop retrieval, and narrative reasoning, MDia improves macro accuracy by 3.6 % over the strongest token-reduction baseline and by 3.1 % over Raw CoT, while reducing generated completion tokens by 71% on average. MDia also identifies full or strong support for 19 machine-sociolinguistic rules, including listener-openness asymmetry, public-dialect asymmetry, weak-speaker teaching, foreign-dialect risk, and route simplicity. The results support a receiver-relative view of efficient reasoning: compact reasoning protocols are useful not because they are universally short but because they preserve the state needed by a particular listener, task, and budget.

1 Main

Large language models (LLMs) are deployed not only as isolated text generators but as communicating components in larger inference systems [21, 23, 5]. Routers select models, agents pass intermediate states, critics verify proposed answers, and tool-use pipelines convert reasoning into structured actions [38, 27, 19, 28]. In such systems, an intermediate trace is not only a human-readable explanation. It is also a message produced by one machine component and interpreted by another under latency, cost, and format constraints. This shift raises the central question of this work: when heterogeneous LLMs exchange compact reasoning protocols, which protocols transfer, which fail, and which measurable social regularities predict their usefulness?

Most successful reasoning methods still rely on human-readable natural language. Chain-of-thought, self-consistency, tree search, and graph-style deliberation improve performance by externalizing intermediate states in prose [35, 33, 39, 3]. Natural language is an effective interface for people, but it can be an inefficient and mismatched interface for machine-to-machine reasoning: explanations often include discourse markers, didactic redundancy, and surface conventions that a listener model may not need. In contrast, indiscriminate

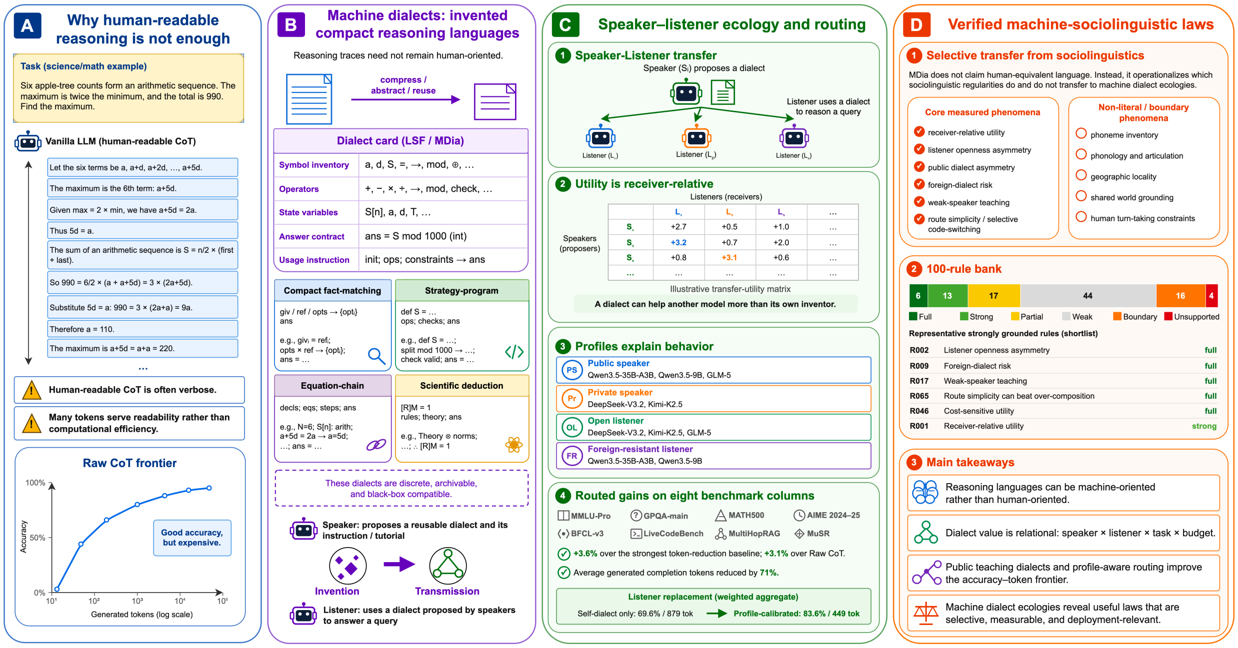

Figure 1: Machine Dialectology: from verbose reasoning traces to a measurable ecology of machine dialects. (A) Human-readable chain-of-thought may devote many generated tokens to explanatory prose, motivating more compact, machine-oriented reasoning representations. (B) Speaker models induce reusable Language Symbolism Frameworks (LSFs), which are discrete, archivable dialect cards that specify symbols, operators, state variables, usage rules and answer contracts. (C) Heterogeneous listener models exhibit receiver-relative and asymmetric gains from these dialects, enabling profile-, task- and budget-aware routing. (D) Aggregating speaker–listener–dialect–task events yields a machine-sociolinguistic rule bank that identifies supported regularities, including listener openness, public-dialect asymmetry and weak-speaker teaching, and delineates where analogies to human language break down.

shortening can remove variables, evidence links, parse commitments, or verification tags that the listener does need. The problem is therefore not simply to "write less", but to choose a representational convention that preserves the right state for a specific listener and task.

Dialectology offers a useful conceptual framework because it treats variation as socially conditioned rather than as noise around a single standard form. Human speakers accommodate audiences, listeners resist unfamiliar styles, public forms spread, private forms remain local, and code-switching occurs when communicative goals change [17, 2, 9, 14]. Computational work on human dialects and language varieties likewise shows that evaluation must account for variation, transfer, and performance disparities between linguistic communities [15, 6, 10]. MDia extends this measurement perspective to model-created symbolic protocols. A *speaker* induces a symbolic protocol; a *listener* executes it; a *dialect* is a reusable Language Symbolism Framework (LSF) with a symbol inventory, grammar, reasoning operators, and answer contract; *publicness* measures whether a dialect helps foreign listeners; *openness* measures whether a listener benefits from foreign dialects; and *code-switching* is the route decision that chooses among dialects under a task and budget.

Existing work establishes several components of this story, but not the full speaker–listener analysis. Concise-thought and sketch-style prompting shorten explicit rationales [20, 36, 1]; program-aided reasoning shifts computation into external code or tools [8, 4]; prompt optimization searches for better instruction strings [37, 7, 40]; and emergent-communication studies show that artificial agents can develop compact conventions under utility and complexity pressures [18, 31, 16]. A parallel line bypasses discrete text: Coconut-style latent reasoning feeds continuous hidden states back into the model, Interlat sends last-layer hidden states between agents, and LatentMAS uses shared latent working memory for multi-agent collaboration [11, 41]. These latent-communication systems show that natural-language traces are not the only possible communication medium. However, they typically require access to hidden representations, layer or architecture alignment, and nonstandard serving interfaces, making them difficult to deploy for black-box API models, heterogeneous commercial systems, or audit logs that must remain stable across model versions. MDia therefore studies the complementary regime of *discrete, archivable, and black-box-compatible* machine dialects that can be produced, interpreted, compared, and routed without reading a model's internal activations.

MDia does not require a human-authored meta-dialect for cold start. Agents first solve sampled queries directly; correct and concise traces are selected as high-leverage evidence; reusable dialect cards are induced or updated from those traces; speaker–listener transfer is measured; and a dynamic archive stores query, benchmark, speaker, listener, dialect, route plan, correctness, token cost, and provenance fields. At inference time, MDia selects dialects from historical records using listener and task profiles, rather than treating model names or raw speaker strength as sufficient descriptions.

The paper makes four contributions. First, it defines machine dialectology as a measurement framework whose unit of analysis is the speaker–listener–dialect–task event. Second, it introduces a dynamic archive and profile-guided routing workflow to select dialects under accuracy and token budgets. Third, it shows that MDia improves the accuracy–token frontier against Raw CoT [35], CoD [36], SoT [1], and CLSR [24]. The primary evidence comes from four strong-reasoning benchmarks: MMLU-Pro, GPQA-main, MATH500, and AIME 2024–25 [34, 25, 12, 32]. Additional format-sensitive benchmarks, including BFCL-v3, LiveCodeBench-output, MultiHopRAG, and MuSR [22, 13, 30, 29], test whether dialect cards preserve schema, parseability, evidence sufficiency, and narrative state. Fourth, it extracts a machine-sociolinguistic rule bank, showing that dialect utility is receiver-relative, that weak speakers can sometimes teach stronger or more verbose listeners, that foreign dialects require negative-transfer guards and that simple profile-aware route choices can outperform over-composed multi-dialect plans when the archive contains high-variance candidates.

2 Results

Figure 1 summarizes the MDia workflow. The central unit is the *dialect event*: a record containing the query, benchmark, split, speaker, listener, dialect card, route decision, generated response, parsed answer, correctness, token cost, and provenance. This event-level archive converts heterogeneous LLM communication into a measurable ecology. The same dialect can function as a private self-talk convention for its speaker, a public teaching language for a weaker listener, or a risky foreign dialect for an expert listener whose native reasoning procedure already contains a richer intermediate structure. By recording these interactions systematically, MDia measures publicness, openness, foreign-dialect resistance, task affinity, overcompression risk, and route cost, rather than relying on anecdotal impressions of individual model families.

2.1 MDia improves the accuracy–token frontier

Table 1 reports the main accuracy–token frontier. Across matched model–benchmark cells, MDia improves the macro-averaged frontier: it gains 3.6 % over the strongest token-reduction baseline selected per matched condition and 3.1 % over Raw CoT, while reducing generated completion tokens by 71% on average. In several near-saturated or format-fragile settings, another baseline may match MDia on raw accuracy, but MDia usually reaches a better operating point once generated-token cost and route stability are considered. The first four benchmark columns show that profile-guided dialects preserve evidence ranking, option elimination, equation state, and answer commitments while removing natural-language redundancy. The last four columns stress different dialect demands: function-call construction requires schema and argument-binding discipline; code-output prediction requires exact output contracts; multi-hop retrieval requires evidence guarding and short verification; and narrative reasoning requires candidate contrast and state consistency.

Frontier interpretation is especially important in near-saturated or resistant-listener settings. On MATH500 and MultiHopRAG, several baseline rows are already close to ceiling, so MDia must preserve decisive verification state while still reducing token cost. On AIME and MuSR, the gains depend more strongly on whether the selected dialect preserves equation states, candidate contrasts, state transitions, and final-answer checks. In BFCL-v3 and LiveCodeBench-output, the final format is part of the reasoning protocol rather

Table 1: Main accuracy–token results. Each entry reports accuracy in percent followed by average generated completion tokens. The MDia row reports the metadata-routed instantiation and is therefore reported simply as MDia. The values are averaged over five test trials. See Appendix A.13 for the details of each benchmark.

Reasoner	Method	MMLU-Pro	GPQA-main	MATH500	AIME 24/25	BFCL-v3	LiveCodeBench	MultiHopRAG	MuSR
Qwen3.5-9B	Raw CoT	74.6 (779)	69.6 (1363)	98.8 (1104)	46.6 (3506)	67.2 (135)	13.1 (341)	92.3 (85)	59.4 (155)
	CoD	73.5 (306)	54.9 (390)	76.6 (382)	24.3 (1684)	34.1 (92)	5.8 (150)	82.1 (58)	42.3 (112)
	SoT	77.1 (473)	56.2 (1436)	88.0 (599)	35.0 (2824)	65.2 (105)	3.5 (202)	84.4 (79)	59.1 (176)
	CLSR	78.3 (572)	56.7 (872)	89.6 (561)	44.2 (2147)	67.0 (56)	10.2 (125)	85.8 (44)	58.5 (68)
	MDia	**80.2 (275)**	**59.3 (439)**	**96.8 (284)**	**49.8 (1072)**	**67.2 (33)**	**13.4 (45)**	**95.4 (18)**	**59.7 (35)**
Qwen3.5-35B-A3B	Raw CoT	84.1 (842)	69.2 (2218)	98.0 (1049)	61.2 (3361)	69.1 (138)	13.5 (352)	92.3 (94)	32.8 (187)
	CoD	79.1 (115)	62.1 (445)	83.2 (358)	48.2 (2159)	54.6 (86)	22.5 (240)	92.3 (79)	58.6 (174)
	SoT	81.1 (547)	64.3 (1620)	91.2 (551)	50.5 (2591)	64.5 (128)	11.8 (237)	92.0 (105)	52.2 (232)
	CLSR	80.6 (933)	65.2 (1769)	95.6 (539)	62.1 (2903)	64.5 (67)	19.5 (172)	93.4 (72)	50.5 (128)
	MDia	**84.8 (103)**	**71.5 (672)**	**99.2 (267)**	**65.4 (1105)**	**69.7 (34)**	**26.5 (79)**	**97.2 (20)**	**54.4 (47)**
Kimi-K2.5	Raw CoT	79.6 (438)	67.9 (1166)	98.4 (493)	71.8 (1912)	61.4 (93)	25.8 (330)	92.5 (118)	42.7 (236)
	CoD	77.6 (175)	65.2 (566)	94.0 (229)	56.3 (1572)	55.3 (102)	24.5 (223)	90.9 (78)	25.6 (175)
	SoT	78.1 (285)	61.6 (763)	96.8 (323)	57.3 (1806)	59.2 (115)	25.5 (244)	84.1 (119)	42.1 (216)
	CLSR	80.6 (278)	67.0 (896)	95.6 (262)	67.0 (2046)	69.2 (105)	32.7 (136)	89.8 (105)	45.7 (126)
	MDia	**82.6 (162)**	**69.7 (485)**	**97.2 (155)**	**69.4 (1238)**	**76.5 (40)**	**32.4 (58)**	**92.5 (34)**	**62.4 (76)**
GLM-5	Raw CoT	77.6 (502)	64.7 (1049)	96.8 (1297)	49.5 (3357)	47.2 (116)	25.8 (228)	84.4 (91)	58.6 (173)
	CoD	76.6 (85)	65.8 (329)	84.2 (258)	29.7 (1072)	44.8 (109)	24.2 (234)	92.0 (69)	57.5 (146)
	SoT	76.1 (218)	67.2 (875)	86.4 (336)	33.0 (1875)	34.6 (112)	25.3 (165)	76.7 (90)	51.2 (151)
	CLSR	78.1 (131)	71.4 (607)	87.2 (335)	38.8 (1648)	45.2 (67)	24.8 (112)	81.5 (72)	54.3 (68)
	MDia	**81.5 (80)**	**72.8 (258)**	**93.9 (164)**	**48.5 (872)**	**48.0 (47)**	**27.5 (34)**	**92.5 (37)**	**61.7 (78)**
DeepSeek-V3.2	Raw CoT	83.1 (529)	67.0 (1382)	98.4 (899)	58.3 (3076)	75.1 (121)	13.0 (329)	92.2 (108)	69.2 (158)
	CoD	78.6 (134)	62.1 (516)	89.2 (273)	40.8 (2133)	68.5 (107)	23.5 (175)	92.0 (52)	42.3 (78)
	SoT	82.6 (260)	68.3 (882)	94.4 (463)	51.5 (2610)	65.4 (115)	13.2 (137)	84.0 (75)	52.0 (106)
	CLSR	85.3 (392)	67.2 (635)	96.4 (551)	54.2 (1816)	67.2 (66)	15.6 (48)	92.0 (52)	52.5 (45)
	MDia	**87.5 (170)**	**69.3 (506)**	**98.8 (225)**	**58.5 (1742)**	**72.7 (38)**	**24.2 (43)**	**98.4 (35)**	**58.6 (49)**

Table 2: Model-profile features used for profile-level interpretation. Hard-math strength averages MATH500 and AIME raw accuracy, knowledge strength averages MMLU-Pro and GPQA-main raw accuracy, and verbosity is the mean raw generated-token count across the four benchmarks.

Model	Hard-math	Knowledge	Verbosity	Profile label
Qwen3.5-9B	72.7	68.3	1892	unstable but dialect-adaptable listener; foreign-resistant listener; public speaker
Qwen3.5-35B-A3B	79.6	76.6	1867	verbose expert reasoner; foreign-resistant listener; public speaker
Kimi-K2.5	81.1	72.4	1001	compact expert reasoner; open listener; private speaker
GLM-5	68.8	69.1	1598	unstable but dialect-adaptable listener; open listener; public speaker
DeepSeek-V3.2	78.1	71.1	1511	compact expert reasoner; open listener; private speaker

than a cosmetic post-processing step. Thus, the main performance claim is not that every compact dialect dominates every raw trace but that metadata- and profile-guided dialect selection provides a better operating point on the accuracy–token frontier than generic short-reasoning prompts. Theorem 1 formalizes this choice of route as a finite-horizon constrained control problem, and Theorem 2 explains why token savings must come from increasing useful task information per generated token rather than from indiscriminate compression.

2.2 Profiles explain dialect behavior better than model-name summaries

Table 2 converts the model identities into interpretable profiles. The profile labels are derived from raw benchmark accuracy, generated-token verbosity, hard-math strength, knowledge strength, self-dialect behavior, foreign openness, foreign resistance, and speaker publicness. The resulting categories include verbose expert reasoners, compact expert reasoners, unstable but dialect-adaptable listeners, open listeners, resistant listeners, public speakers, and private speakers.

This profile view changes the way the results should be interpreted. The important finding is not that a particular LLM has a fixed personality; instead, model families occupy different roles in the dialect ecology because they differ in verbosity, raw capability, schema discipline, openness to foreign dialects and ability to produce public teaching dialects. Compact expert listeners often benefit from dialects that preserve decisive

Table 3: Teaching-advantage examples from validation-style transfer analysis. Teaching advantage is the improvement of a speaker family for foreign listeners relative to the corresponding self-dialect baseline.

Benchmark	Speaker family	Teaching advantage	Profile interpretation
GPQA-main	Qwen3.5-35B-A3B	+14.17	verbose expert reasoner; foreign-resistant listener; public speaker
GPQA-main	Qwen3.5-9B	+11.38	unstable but dialect-adaptable listener; foreign-resistant listener; public speaker
MATH500	GLM-5	+8.70	unstable but dialect-adaptable listener; open listener; public speaker
MATH500	Qwen3.5-9B	+8.50	unstable but dialect-adaptable listener; foreign-resistant listener; public speaker
MMLU-Pro	Qwen3.5-35B-A3B	+5.60	verbose expert reasoner; foreign-resistant listener; public speaker
GPQA-main	DeepSeek-V3.2	+4.91	compact expert reasoner; open listener; private speaker
AIME	Qwen3.5-9B	+4.37	unstable but dialect-adaptable listener; foreign-resistant listener; public speaker
AIME	Qwen3.5-35B-A3B	+3.40	verbose expert reasoner; foreign-resistant listener; public speaker
MMLU-Pro	Qwen3.5-9B	+2.36	unstable but dialect-adaptable listener; foreign-resistant listener; public speaker
AIME	GLM-5	+1.46	unstable but dialect-adaptable listener; open listener; public speaker

Table 4: Listener replacement with fixed dialect archive. Each entry reports accuracy in percent followed by average generated completion tokens. Profile-calibrated routing consistently improves over self-dialect and name-based routing, supporting the view that dialect utility depends on speaker–listener compatibility rather than model identity alone.

Listener	Name-based	Self-dialect only	Profile-calibrated	Raw-score only
DeepSeek-V3.2	74.6 (929)	74.6 (929)	**87.6** (563)	**87.6** (563)
GLM-5	63.0 (498)	63.0 (498)	**81.1** (262)	**81.1** (262)
Kimi-K2.5	73.4 (679)	73.4 (679)	**89.1** (352)	**89.1** (352)
Qwen3.5-35B-A3B	69.5 (968)	69.5 (968)	**87.3** (577)	**87.3** (577)
Qwen3.5-9B	61.8 (984)	67.5 (1321)	72.8 (491)	**82.5** (338)†
Weighted mean	68.5 (812)	69.6 (879)	**83.6** (449)	85.7 (423)†

†The raw-score policy has slightly smaller held-out coverage than the other policies and is therefore used as a coverage-qualified comparator.

symbolic states without excessive narration. Verbose expert reasoners can use public dialects as a way to eliminate redundant prose while retaining the verification structure. Unstable but dialect-adaptable listeners can be helped by public teaching dialects, especially when the dialect gives a short schema for selecting evidence or maintaining the equation state.

Table 3 gives representative teaching-advantage examples. The metric compares how useful a speaker family is for foreign listeners relative to its own self-dialect baseline. Positive values indicate that a dialect family functions more like a public teaching language than a private self-talk code. This is the most direct evidence that the dialect value is receiver-relative. A dialect may be valuable because it is easy for others to adopt, not because it is the most faithful expression of its speaker's own reasoning procedure.

2.3 Dialect utility transfers across listener families

A central claim of MDia is that a useful machine dialect is relational: it is not intrinsically strong or weak but useful for a particular listener under a particular task distribution. We test this claim by replacing the listener while keeping the dialect archive fixed. Four routing policies are compared. The self-dialect policy restricts the listener to dialects produced by its own family. The name-based policy uses the model identity as the primary routing cue. The profile-calibrated policy selects routes from held-out validation estimates of accuracy and token cost. The raw-score policy selects high-scoring archive routes without the full listener-profile calibration.

Table 4 shows the result with 64 calibration examples per listener. Profile-calibrated routing substantially outperforms self-dialect routing for every listener family. On the weighted aggregate, the accuracy improves from 69.6% to 83.6%, while the average generated tokens decrease from 879 to 449. This shows that the archive contains public dialects whose value is not reducible to self-talk. A dialect produced by one model family can become a better teaching language for another family than the listener's own native dialect.

Table 5: Main machine-sociolinguistic rule candidates. The full 100-rule bank is reported in Supplementary Table 22 and analyzed rule-by-rule in Appendix A.9.

ID	Rule	Score	Support	Operational implication
R002	Listener openness asymmetry	1.00	full	Condition route scores on listener profile and parse risk.
R016	Public dialect asymmetry	1.00	full	Prefer public/teaching speakers over raw-score-only speaker selection.
R065	Route simplicity can beat over-composition	1.00	full	Use profile- and task-aware route selection with negative-transfer guards.
R009	Foreign-dialect risk	0.93	full	Condition route scores on listener profile and parse risk.
R017	Weak-speaker teaching	0.90	full	Prefer public/teaching speakers over raw-score-only speaker selection.
R046	Cost-sensitive utility	0.90	full	Optimize accuracy-token utility, not compression alone.
R035	Tool-call schema-binding pressure	0.78	strong	Use task-family-specific dialect pools and route weights.
R069	Oracle gap reveals unused social information	0.75	strong	Use profile- and task-aware route selection with negative-transfer guards.
R100	Negative-transfer guards are essential for safe deployment	0.75	strong	Log model versions, seeds, cache costs, and failure diagnostics before deployment.
R001	Receiver-relative utility	0.71	strong	Condition route scores on listener profile and parse risk.
R061	Pragmatic code-switching	0.50	partial	Use profile- and task-aware route selection with negative-transfer guards.

Table 6: Support distribution for the 100-rule bank. Full and strong rules are used as the core empirical mechanism claims; partial, weak, boundary and unsupported rules are retained as constraints and future-work diagnostics.

Support level	full	strong	partial	weak	boundary	unsupported
Count	6	13	17	44	16	4

The result gives a concrete operational meaning to the machine dialectology. If dialects were merely private abbreviations, self-dialect routing should have dominated. Instead, profile-calibrated public dialects often transfer better than self-dialects. MDia therefore studies not only whether a compact symbolic trace is accurate, but also who can read it, who benefits from it, and under which task distribution it should be selected. This is the machine analog of audience design and code-switching: the best form is determined jointly by speaker, listener, and communicative goal.

2.4 Machine-sociolinguistic rules summarize the mechanism

The rule bank expands the mechanism analysis from a small illustrative list to 100 testable machine-sociolinguistic rules. The purpose of the bank is not to make every sociolinguistic analogy true. Instead, each rule receives an operational metric, a support score, and a support label, so that strongly supported rules can shape the main claims while weak, boundary, and unsupported rules delimit where the analogy fails. The support distribution in Table 6 is therefore part of the evidence: only 6 rules are fully supported and 13 are strongly supported, while many remain weak or boundary-level constraints on the analogy.

Several representative rules are central to our work; see Appendix A.21 for details. *Listener openness asymmetry* and *public dialect asymmetry* show that speakers and listeners are not interchangeable endpoints. Some dialect families are broadly adoptable, whereas some listeners benefit substantially from foreign dialects, and others resist them. *Receiver-relative utility* is the general form of this observation: a dialect cannot be ranked without specifying the listener and task distribution. This directly explains why profile-guided routing is more informative than model-name summaries.

Weak-speaker teaching is the most surprising rule. A lower raw-accuracy speaker can produce a dialect that helps foreign listeners because teaching value depends on clarity, redundancy, answer-contract stability, and compatibility with the listener's inductive biases, not only on the speaker's own final-answer accuracy. This

Table 7: Largest fixed-archive cross-generational functional influence effects. Values are leave-one-source-out decreases in downstream accuracy when a source dialect family is removed from the pooled archive.

Benchmark	Removed source family	Δ acc.	Std.	Conditions
AIME	Qwen3.5-35B-A3B	+4.85	4.31	15
GPQA-main	Qwen3.5-9B	+4.64	4.03	15
AIME	DeepSeek-V3.2	+4.47	2.74	15
AIME	GLM-5	+3.82	1.96	15
AIME	Kimi-K2.5	+3.62	2.58	15
GPQA-main	DeepSeek-V3.2	+2.50	2.60	15
GPQA-main	Kimi-K2.5	+2.29	2.19	15
AIME	Qwen3.5-9B	+2.20	5.76	15
GPQA-main	GLM-5	+2.14	2.60	15
MATH500	DeepSeek-V3.2	+1.68	1.20	15
MMLU-Pro	DeepSeek-V3.2	+1.53	1.12	15
MATH500	Qwen3.5-35B-A3B	+1.47	1.60	15

rule is visible in the teaching-advantage examples of Table 3 and is one reason why MDia maintains an archive of speaker roles rather than discarding all non-frontier speakers.

Foreign-dialect risk and *negative-transfer guards* are the safety rules of the system. They prevent MDia from becoming a naive dialect ensemble. A foreign dialect can be harmful when it over-compresses evidence, imposes an unfamiliar answer format, or suppresses the verification steps needed by a strong listener. The route-repair analysis in Appendix A.2 shows that more composition is not automatically better; route simplicity can beat over-composition when the archive contains high-variance candidates. Thus, the most robust MDia policy is not maximally social, but selectively social.

Cost-sensitive utility links the rule bank back to the main performance table. A dialect is useful when its correctness gain, token cost, parse risk, and listener resistance jointly improve the accuracy–token frontier. This is why the route score in Methods includes both accuracy and cost terms, and why a near-saturated benchmark can still support MDia when the dialect removes a large amount of generated natural-language reasoning without materially damaging correctness. The appendix gives a rule-by-rule analysis of all 100 rules in Appendix A.9; the figures and narratives there are referenced here because the rule bank is a central contribution rather than a mere supplementary list.

2.5 Cross-generational influence is directed and task-dependent

Table 7 reports the fixed-archive leave-one-source functional influence. For a target listener and benchmark, we compare a pooled dialect archive with a counterfactual archive that removes one source dialect family. A positive value means that removing the source family reduces downstream accuracy. The largest effects occur on AIME and GPQA-main, indicating that hard mathematical verification and scientific discrimination are especially sensitive to which dialect families seed the archive.

This analysis measures the functional influence in saved dialect records rather than parameter updating. Within that scope, it is informative because it separates *being a strong solver* from *leaving useful conventions in the archive.* AIME favors source families that contribute reusable strategy operators, equation bindings, and verification tags. GPQA-main favors dialect families that preserve evidence ranking and option elimination. MATH500 and MMLU-Pro show smaller top effects, which is consistent with the ceiling and task-headroom rules in the rule bank. The result gives a machine-sociolinguistic interpretation of cross-generation effects: dialect families survive not because they are universally strong but because they provide useful conventions for particular listener and task profiles.

3 Discussion

Machine dialectology as a measurement framework. MDia changes the unit of analysis from an individual prompt to a speaker–listener–dialect–task event. This matters because efficient reasoning in heterogeneous systems is not a property of a string alone; it is a relation among the model that produced the protocol, the model that interprets it, the task it serves, and the cost the system can afford. The dynamic archive gives this relation a measurable form, allowing publicness, openness, resistance, teaching, and code-switching to be studied as empirical variables rather than as informal metaphors.

Receiver-relative intelligence. The results complicate ordinary model leaderboards. A model can be a strong direct reasoner but a poor public dialect speaker; another can be weaker as a direct solver but useful as a teacher because its dialect is easier for foreign listeners to adopt. This does not make the raw accuracy irrelevant; it makes the raw accuracy coordinate in a richer communicative profile. For deployment, the useful question is not only "which model is best?", but "which model's dialect should this listener use on this task under this budget?"

Accuracy–token frontiers, not compression alone. MDia differs from prompt compression because it optimizes which task state survives compression. CoD and SoT reduce the amount of text, but do not explicitly model whether a listener needs variable bindings, evidence ranks, verification markers, parse contracts, or a more redundant public teaching dialect. The main table therefore supports a frontier claim: MDia improves accuracy relative to short-reasoning baselines, sharply reduces completion tokens relative to Raw CoT, and reveals where near-ceiling tasks or resistant listeners make the frontier trade-off explicit.

Code-switching rather than indiscriminate ensembling. route-repair and rule-bank analyzes show that routing is not simply a larger ensemble. Unconstrained composition can import negative transfer and inflate cost. The stronger policy is pragmatic: use a compact public dialect when the listener is open and the task is schema-like; fall back to self-dialect or Raw reasoning when the listener is resistant; invoke verifier-rich dialects when hard mathematics requires explicit checks; and avoid ultra-compact dialects for compression-fragile listeners. This pattern is closest to code-switching under communicative constraints.

Interpretability and auditability. Compact machine dialects are less directly readable than ordinary natural-language rationales and should not be presented as transparent explanations of internal model causality. They are algorithmic artifacts that are used to improve prediction in a cost budget. At the same time, dialect cards are more structured than unconstrained prose: auditors can inspect variable bindings, answer contracts, verification tags, parse failures, and route decisions. For high-stakes deployment, MDia should store the dialect card, selected route, raw response, parsed response, and a human-readable final explanation.

Boundaries and future work. Additional work should examine broader task families, multimodal dialects, retrieval and tool-use dialects, human–machine accommodation, API drift, standardized machine-dialectology benchmarks, and safety constraints for opaque compact traces. The central claim remains conservative: MDia does not claim that LLMs spontaneously develop human-equivalent languages, but it shows that heterogeneous model societies exhibit measurable, useful, and failure-prone dialect-like communication patterns.

4 Methods

4.1 Overview of the MDia pipeline

MDia is a test-time framework with eight stages. First, it collects direct responses from a heterogeneous model population on training or validation examples. Second, it evaluates those responses and selects high-leverage traces that are both correct and token-efficient. Third, it induces or updates dialect cards from the selected traces, preserving symbol inventories, schemas, reasoning operators, and answer contracts. Fourth, it evaluates speaker–listener transfer by asking listener models to solve held-out examples under speaker dialects. Fifth, it builds a dynamic archive of dialect events and profile features. Sixth, it validates machine-sociolinguistic rules in archive statistics. Seventh, it routes dialects at test time using profile, task, and cost information estimated without the target answer. Eighth, it evaluates final answers on held-out examples with matched evaluators and generated-token accounting. In general, MDia can be interpreted as both a data protocol for measuring communication among models and an optimization procedure over an accuracy–cost frontier.

4.2 Benchmarks and partitions

We evaluated on eight benchmarks: MMLU-Pro, GPQA-main, MATH500, AIME 2024–25, BFCL-v3, LiveCodeBench-output, MultiHopRAG and MuSR [34, 26, 12, 32, 22, 13, 30, 29]. MMLU-Pro measures broad professional and factual reasoning. GPQA-main measures expert-level scientific discrimination. MATH500 measures competition-style mathematical problem solving. AIME measures hard contest mathematics with short final answers. BFCL-v3 measures the construction of functions under schema constraints. LiveCodeBench-output measures exact stdout prediction for programming-problem inputs. MultiHopRAG measures evidence-based multihop question answering. MuSR measures multistep soft reasoning over natural-language narratives.

Let $\mathcal{X}_b$ be the query space and $\mathcal{Y}_b$ be the evaluator answer space for benchmark b. The dataset is

$$\mathcal{D}^{(b)} = \{(x_n^{(b)}, y_n^{(b)})\}_{n=1}^{N_b}, \qquad x_n^{(b)} \in \mathcal{X}_b, \quad y_n^{(b)} \in \mathcal{Y}_b. \tag{1}$$

A benchmark evaluator is a mapping $\mathcal{E}_b : \mathcal{R} \to \mathcal{Y}_b \cup \{\perp\}$ from a text response $r \in \mathcal{R}$ to a canonical response or a parse failure $\perp$. Training or evolution examples are used to generate candidate traces and dialect cards. Validation-style examples are used to estimate profiles, transfer matrices, rule support, and route-selection statistics. Held-out reporting examples are used for Table 1. Each query also carries an observable metadata vector $g_b(x) \in \mathcal{G}_b$ that is available before answer generation. For the four format-sensitive benchmark families, $g_b(x)$ contains the benchmark tag for BFCL-v3 and LiveCodeBench-output, the question type for MultiHopRAG, and the narrative subdomain for MuSR. The route policy may use observable benchmark family, answer-format and query-subtype metadata available before generation, but never uses gold answers, hidden labels or feedback from a candidate answer when selecting a dialect. The speaker–listener matrices, rule validation and cross-generational influence analyzes should therefore be interpreted as mechanistic validation evidence unless explicitly marked as held-out evaluations.

4.3 Model population and profile features

Let $\mathcal{M} = \{M_1, \ldots, M_L\}$ be a population of frozen LLM agents. Each model is treated as a stochastic text transducer $M_i : \mathcal{X}_b \times \mathcal{D}_{\text{card}} \times \mathcal{S} \to \Delta(\mathcal{R})$, where $\mathcal{D}_{\text{card}}$ is the dialect-card space, $\mathcal{S}$ is the decoding-state space and $\Delta(\mathcal{R})$ is the set of distributions over text responses. In the reported experiments, the population contains Qwen3.5-9B, Qwen3.5-35B-A3B, Kimi-K2.5, GLM-5 and DeepSeek-V3.2. A model can act as a direct solver, dialect speaker, dialect listener, router, or judge depending on the stage. The model weights are not updated.

Profiles are computed from validation and audit records rather than from model names. For model M_i, the profile vector is $\phi_i \in \mathbb{R}^p$ and contains raw benchmark accuracies, raw generated-token statistics, hard-math strength, knowledge strength, verbosity, self-dialect strength, foreign openness, foreign resistance, speaker publicity, and teaching advantage. These features are used to describe the results at the level of model types rather than isolated product names.

4.4 Dialect representation

A machine dialect is stored as a persistent Language Symbolism Framework card. For speaker M_j, benchmark b and evolution state t, the card is represented as

$$D_j^{(b,t)} = \left(\mathcal{V}_j^{(b,t)}, \mathcal{G}_j^{(b,t)}, \mathcal{O}_j^{(b,t)}, \mathcal{R}_j^{(b,t)}, \rho_j^{(b,t)}\right) \in \mathcal{D}_{\text{card}}. \tag{2}$$

Here $\mathcal{V} \subset \Sigma^*$ is a finite symbol inventory in the token alphabet Σ, $\mathcal{G}$ is a grammar or output schema, $\mathcal{O}$ is a finite set of reasoning operators, $\mathcal{R}$ is a set of validity and usage rules, and $\rho \in \mathbb{R}^q$ is the empirical dialect profile. The profile stores validation accuracy, generated-token statistics, success domains, failure modes, and listener-specific compatibility estimates. The card is textual because the agents are language models, but it is not a one-time prompt; it is reused across queries and evaluated as a persistent protocol.

4.5 Direct-response bootstrapping and high-leverage selection

MDia starts from direct model responses rather than from a human-authored meta-dialect. Let $\mathrm{Tok} : \mathcal{R} \to \mathbb{N}$ return the generated completion-token count. In the initial state, each model directly answers the sampled examples:

$$\begin{aligned} r_{j,n}^{(b,0)} &\sim M_j(x_n^{(b)}; \varnothing), \qquad r_{j,n}^{(b,0)} \in \mathcal{R}, \\ z_{j,n}^{(b,0)} &= \mathbf{1}[\mathcal{E}_b(r_{j,n}^{(b,0)}) = y_n^{(b)}], \qquad c_{j,n}^{(b,0)} = \mathrm{Tok}(r_{j,n}^{(b,0)}). \end{aligned} \tag{3}$$

Correct and concise traces expose native reasoning habits that can seed a dialect. In the evolution state t, a dialect-conditioned response is

$$r_{j,n}^{(b,t)} \sim M_j(x_n^{(b)}; D_j^{(b,t)}). \tag{4}$$

For each query, the correct-response set is

$$\mathcal{C}_n^{(b,t)} = \{(j, r_{j,n}^{(b,t)}, c_{j,n}^{(b,t)}) : \mathcal{E}_b(r_{j,n}^{(b,t)}) = y_n^{(b)},\ c_{j,n}^{(b,t)} = \mathrm{Tok}(r_{j,n}^{(b,t)})\}. \tag{5}$$

The high-leverage set is the token-efficient subset of correct traces:

$$\mathcal{H}_n^{(b,t)} = \mathrm{TopK}_{(j,r,c)\in\mathcal{C}_n^{(b,t)}}[-c], \qquad \mathcal{H}^{(b,t)} = \bigcup_n \mathcal{H}_n^{(b,t)}. \tag{6}$$

This selection pressure favors traces that communicate task-relevant state with fewer generated tokens. A diversity constraint can be used to prevent one high-accuracy speaker from monopolizing the archive.

4.6 Dialect update and dynamic archive

The selected traces are organized by source speaker and exposed to the agent population through a fixed synthesis interface. The discussion context is

$$\Gamma^{(b,t)} = \mathrm{Discuss}(\mathcal{H}^{(b,t)}, \mathcal{M}) \in \mathcal{R}^*. \tag{7}$$

Each model updates its dialect by

$$D_i^{(b,t+1)} = \mathcal{U}_i\left(D_i^{(b,t)}, \mathcal{H}^{(b,t)}, \Gamma^{(b,t)}\right), \qquad \mathcal{U}_i : \mathcal{D}_{\mathrm{card}} \times 2^{\mathcal{R}} \times \mathcal{R}^* \to \mathcal{D}_{\mathrm{card}}. \tag{8}$$

The update permits vertical inheritance from a model's prior dialect and horizontal borrowing from other speakers.

The archive stores one normalized record for every query-dialect execution. Each record contains the query identifier, benchmark, split, speaker model, listener model, speaker profile, listener profile, dialect identifier, route plan, response, parsed answer, gold answer, correctness, prompt tokens, completion tokens, router tokens, latency, estimated cost, seed and source path. This schema is intentionally richer than a final prediction table because MDia requires historical evidence for future route selection.

4.7 Metadata-conditioned dialect controller

The main-table MDia policy is a routed dialect controller. It first chooses an internal dialect family, then asks the listener model to solve the query under that dialect and emit only the required answer object. Formally, let $\mathcal{P}_{\mathrm{meta}}$ be the class of route policies that map observable metadata, listener profile and budget to a dialect controller,

$$\pi : \mathcal{G}_b \times \mathbb{R}^p \times \mathbb{R}_{\geq 0} \to \mathcal{D}_{\mathrm{ctrl}}, \qquad D_{i,n}^{\star} = \pi(g_b(x_n^{(b)}), \phi_i, B). \tag{9}$$

Here $\mathcal{D}_{\mathrm{ctrl}}$ is the finite set of controller-level dialect families and B is the completion-token budget. The selected controller is not a separate method; it is the concrete route policy used by MDia when benchmark metadata is informative enough to choose among dialect families before generation.

Table 8: MDia route-policy components. These components share the same archive schema but differ in how candidate dialects are retrieved, scored, composed and updated. They are components or instantiations of MDia rather than separate methods.

Component	Route policy	Intended use
Archive retrieval	Retrieve historical dialect records by task, query features, listener profile and cost budget.	Low-latency archive lookup.
Profile scoring	Choose candidate dialects using profile features rather than literal model names.	Generalization across model families.
Metadata-conditioned routing	Select a dialect family from observable benchmark or subtype metadata before solving.	Heterogeneous task families with distinct answer contracts.
Split-validation selection	Freeze route selection on a validation partition and evaluate on a held-out partition.	Leakage-controlled validation of dynamic routing.
Dialect composition	Compose planner, solver and verifier dialects when one dialect lacks sufficient coverage.	Hard multi-step reasoning.
Bridge dialects	Route through an intermediate public dialect when a direct foreign dialect is resisted.	Cross-profile accommodation.
Rule-aware penalties	Penalize choices that violate validated machine-sociolinguistic rules.	Safer routing under limited labels.
Cost-aware utility	Optimize accuracy subject to generated-token, latency and monetary-cost budgets.	Deployment-time budget control.
Answer-contract guard	Emit only the minimal parseable answer object required by the evaluator.	Function calls, exact stdout, short-answer QA and multiple choice.

The controller separates *internal dialect* from *external answer contract*. For a benchmark b, the answer contract is a parser-facing schema $\kappa_b \in \mathcal{K}$. BFCL-v3 uses a tool-call list, LiveCodeBench-output uses an exact-stdout field, MultiHopRAG uses a short-answer field, and MuSR uses a multiple-choice answer index plus answer text. The solver may use a schema, contrastive, verifier or evidence-guard dialect internally, but the emitted response must satisfy κ_b without visible rationale. This design keeps completion-token cost low while preserving the variables, bindings and checks that the evaluator actually tests.

Table 9: Metadata-conditioned MDia route map. All route keys are observable before answer generation. Routes may use benchmark family, answer-format and query-subtype metadata available before generation; no route uses gold answers, hidden labels or candidate-answer feedback.

Benchmark family	Observable route key	Dialect controller	Internal constraint
MMLU-Pro, GPQA-main, MATH500, AIME	Benchmark tag; answer format; listener profile; budget.	Profile-utility LSF route.	Select the highest-utility dialect card under Eq. 14; preserve option, equation and answer commitments.
BFCL-v3	Function-call benchmark route.	Schema-zip function dialect.	Bind exact function names, required arguments and schema defaults; split parallel “respectively” requests into aligned calls.
LiveCodeBench-output	Output-prediction benchmark route.	Schema-first stdout dialect.	Simulate the supplied input silently and emit the exact normalized stdout string.
MultiHopRAG	`comparison_query`.	Direct evidence dialect.	Bridge named entities across provided evidence and answer only from supported facts.
MultiHopRAG	`inference_query`.	Short verifier dialect.	Produce a silent candidate, verify it against all evidence facts, then emit the shortest supported answer.
MultiHopRAG	`null_query`.	Null-aware guard dialect.	Return an insufficiency answer when support is missing rather than over-answering.
MuSR	`murder_mystery`; `object_placements`.	Contrastive state-consistency dialect.	Track candidates, states, observations and contradictions; reject choices that violate any constraint.
MuSR	`team_allocation`.	Schema allocation dialect.	Represent people, roles and compatibility constraints in a compact schema before choosing the final option.

The same principle covers both broad strong-reasoning columns and format-sensitive columns. For broad QA and mathematics, MDia mainly selects among evolved LSF cards using listener profiles and cost-aware utility. For function calls, code-output prediction, evidence QA and narrative reasoning, the benchmark metadata identifies which symbolic variables are fragile: tool names and argument defaults, stdout formatting, evidence sufficiency, or candidate-state consistency. The route map in Table 9 therefore makes the dialect choice explicit without turning MDia into a list of unrelated prompts.

4.8 Speaker-listener transfer

To measure transfer, we separate the speaker that generated a dialect from the listener that executes it. Listener M_i solves held-out query $x_n^{(b)}$ using speaker M_j's dialect:

$$r_{i \leftarrow j,n}^{(b,t)} \sim M_i(x_n^{(b)}; D_j^{(b,t)}). \tag{10}$$

The transfer accuracy and token cost are

$$\begin{aligned} A_{i \leftarrow j}^{(b,t)} &= \frac{1}{|\mathcal{T}^{(b)}|} \sum_{n \in \mathcal{T}^{(b)}} \mathbf{1}[\mathcal{E}_b(r_{i \leftarrow j,n}^{(b,t)}) = y_n^{(b)}], \\ C_{i \leftarrow j}^{(b,t)} &= \frac{1}{|\mathcal{T}^{(b)}|} \sum_{n \in \mathcal{T}^{(b)}} \mathrm{Tok}(r_{i \leftarrow j,n}^{(b,t)}). \end{aligned} \tag{11}$$

The transfer matrix is

$$\mathbf{T}^{(b,t)} = \Big[(A_{i \leftarrow j}^{(b,t)}, C_{i \leftarrow j}^{(b,t)}) \Big]_{i,j=1}^{L} \in ([0,1] \times \mathbb{R}_{\geq 0})^{L \times L}. \tag{12}$$

Diagonal entries are self-dialect use, and off-diagonal entries are foreign-dialect use. These matrices support the social-role metrics and rule validation, but they are not directly averaged with the route-based held-out results in Table 1.

4.9 Routing objective and token accounting

For an MDia inference trace with router messages $q_t \in \mathcal{R}$, dialect-conditioned solver responses $r_{t,k} \in \mathcal{R}$ and aggregation outputs $a_t \in \mathcal{R}$, generated-token cost is

$$C = \sum_{t=1}^{T} \left(\mathrm{Tok}(q_t) + \sum_{k \in I_t} \mathrm{Tok}(r_{t,k}) + \mathrm{Tok}(a_t) \right), \tag{13}$$

where $I_t \subseteq \{1, \ldots, K\}$ is the set of dialects invoked in round t. Generated tokens include router planning, solver calls, intermediate dialect messages and aggregation. Reusable dialect-card input tokens are not included in the main latency-oriented completion-token metric because they can be cached as fixed prompt prefixes, although the raw records preserve prompt-token metadata for cache-aware accounting.

The profile-guided route score is an archive utility rather than an additional trained neural router:

$$\begin{aligned} U : \mathcal{X}_b \times \mathcal{M} \times \mathcal{D}_{\text{card}} &\to \mathbb{R}, \\ U(x, M_i, D_j) = \widehat{p}_{\text{corr}}(x, M_i, D_j) &- \lambda_{\text{tok}} \widehat{C}(x, M_i, D_j) - \lambda_{\text{parse}} \widehat{F}_{\text{parse}}(M_i, D_j) \\ &+ \lambda_{\text{pub}} \widehat{P}(D_j) + \lambda_{\text{task}} \widehat{Q}(D_j, b) - \lambda_{\text{resist}} \widehat{R}(M_i, D_j), \end{aligned} \tag{14}$$

where $\widehat{p}_{\text{corr}}$ is validation-estimated correctness, $\widehat{C}$ is expected generated-token cost, $\widehat{F}_{\text{parse}}$ is parse-failure risk, $\widehat{P}$ is dialect publicness, $\widehat{Q}$ is task affinity and $\widehat{R}$ is listener resistance. A split-validation instantiation freezes this score on a route-selection partition and evaluates the chosen route on a disjoint held-out partition; this is an MDia validation protocol, not a separate method. When the metadata-conditioned controller in Eq. 9 is used, the selected policy is constrained to depend only on $g_b(x)$, ϕ_i and the budget B. The route is accepted for reporting only when its validation estimate improves the accuracy–cost frontier against the matched baselines; after selection, the route map is held fixed during the averaged test trials reported in Table 1.

For a held-out set $\mathcal{T}^{(b)}$, reported accuracy and token cost are

$$A = \frac{1}{|\mathcal{T}^{(b)}|} \sum_{n \in \mathcal{T}^{(b)}} \mathbf{1}[\mathcal{E}_b(r_n) = y_n], \qquad \bar{C} = \frac{1}{|\mathcal{T}^{(b)}|} \sum_{n \in \mathcal{T}^{(b)}} C_n. \tag{15}$$

Aggregate statements are macro-averages over matched model–benchmark conditions unless otherwise specified.

4.10 Rule validation

A machine-sociolinguistic rule R_m is a testable statement about how dialect utility changes with speaker profile, listener profile, task family, token cost, or archive history. Each rule is assigned a metric $s_m \in [0, 1]$ and a support label. Scores close to one indicate that the archived validation evidence matches the rule. Scores close to zero indicate limited or boundary evidence. The complete rule bank is intentionally retained in the supplementary section A.9 because the boundary-case rules are useful for refining the theory even when they are not the main performance claims.

4.11 Statistical and provenance controls

All table values are assembled from saved outputs with canonical evaluator counts and source paths. The main table uses the held-out test records selected under the canonical output audit. Transfer matrices, rule validation, and cross-generational influence are validation-style mechanism analyzes rather than final route evaluations. This separation prevents single-dialect transfer entries from being compared as if they were route-based test-set predictions.

5 Data availability

The benchmarks used in the manuscript are publicly available through their original releases or benchmark repositories: MMLU-Pro, GPQA-main, MATH500, AIME 2024–25, BFCL-v3, LiveCodeBench, MultiHopRAG, and MuSR. The normalized prediction archive, route records, rule-validation tables, and manuscript export tables are organized so that each reported number can be traced to a saved source path.

6 Code availability

Code, prompt templates, dialect-evolution scripts, transfer evaluation, routing scripts, and token-accounting utilities are released with the project repository at `https://github.com/pzqpzq/LSF_MDia`. The release includes scripts to regenerate the main tables from saved records and to reproduce route-selection analyzes under fixed policies.

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A Supplementary Information

A.1 A compact theory of token-efficient dialect routing

For a fixed test query x, an interactive inference policy π induces a multi-round transcript $\mathcal{T}$ and a final prediction $\hat{Y}$. We evaluate the policy by expected token cost $J_C(\pi) = \mathbb{E}[|\mathcal{T}|]$ and accuracy $J_A(\pi) = \Pr(\hat{Y} = Y)$. The optimal accuracy under budget B is

$$A^*(B) = \sup_{\pi: J_C(\pi) \leq B} J_A(\pi). \tag{16}$$

Theorem 1 (Existence of an optimal finite-horizon dialect-routing policy)**.** *Assume a finite dialect pool, an effectively bounded horizon through a stopping action, nonnegative integrable per-step token cost and a well-defined stochastic decoding kernel for each model-dialect-history state. For every $B \geq 0$, the supremum in Eq. 16 is attained by a possibly randomized policy. For every Lagrange multiplier $\lambda \geq 0$, an optimal deterministic finite-horizon policy exists for $\mathbb{E}[\mathbf{1}\{\hat{Y} = Y\} - \lambda|\mathcal{T}|]$.*

The theorem formalizes why routing can be treated as a constrained stochastic-control problem. The dialect pool supplies actions, the transcript supplies writable memory, and the stopping decision controls the accuracy-cost frontier.

Theorem 2 (Information lower bound for generated dialect tokens)**.** *Let Y_x be the finite set of effective answers induced by the evaluator for query x. If a policy achieves target accuracy α with $\delta = 1 - \alpha$, define*

$$I_{\text{req}}(x, \delta) = H(Y \mid X = x) - h_2(\delta) - \delta \log_2(|Y_x| - 1). \tag{17}$$

Let $\kappa_\theta(x)$ upper-bound the conditional task-relevant information revealed by one active generated token under the allowed dialect protocols. Then

$$\mathbb{E}_\pi[|\mathcal{T}| \mid X = x] \geq \frac{\max\{I_{\text{req}}(x, \delta), 0\}}{\kappa_\theta(x)}. \tag{18}$$

Equation 18 does not say that fewer tokens are always better. It says that efficient reasoning requires increasing the amount of useful task state carried per generated token. MDia attempts to increase $\kappa_\theta(x)$ by selecting a dialect that matches the listener and task.

A.2 Split-validation route-selection analysis

This section evaluates a conservative instantiation of MDia in which route selection is frozen on a validation partition and evaluated on a disjoint held-out partition. It documents leakage control for archive-based routing; the method name remains MDia throughout the manuscript. The analysis is supplementary to the main held-out comparison in Table 1.

Table 10: Per-listener held-out MDia split-validation diagnostic. Route selection is frozen using only router-validation records. Accuracy is percent; tokens are average generated completion tokens.

Benchmark	Listener	Raw acc.	Raw tok.	MDia acc.	MDia tok.
MMLU-Pro	Qwen3.5-9B	76.0	847.5	88.0	297.3
MMLU-Pro	Qwen3.5-35B-A3B	86.0	905.4	91.0	304.1
MMLU-Pro	Kimi-K2.5	84.0	439.1	95.0	182.9
MMLU-Pro	GLM-5	78.0	554.8	91.0	118.0
MMLU-Pro	DeepSeek-V3.2	81.0	548.7	92.0	173.2
GPQA-main	Qwen3.5-9B	60.9	2309.3	46.4	1632.6
GPQA-main	Qwen3.5-35B-A3B	69.1	2351.1	80.9	957.7

Table 11: Route-repair diagnostic. Naive full multi-dialect composition can underperform Raw CoT, motivating route simplicity and negative-transfer guards in MDia.

Strategy	Accuracy	Completion tokens	Records
Full MDia routing with multi-dialect composition	62.04	464.2	30
Globally best dialect by cached utility	71.70	892.0	30
Oracle upper bound from cached predictions	86.72	582.4	30
Random foreign dialect (20 seeds)	65.87	769.6	30
Raw	75.49	1105.0	30
Self-dialect only	66.67	795.5	30
Social-aware single-dialect routing	71.09	859.4	30
Task-aware routing without speaker–listener profile	71.70	892.0	30

Benchmark	Listener	Raw acc.	Raw tok.	MDia acc.	MDia tok.
GPQA-main	Kimi-K2.5	69.1	1242.5	83.6	517.0
GPQA-main	GLM-5	62.7	1259.0	73.6	277.0
GPQA-main	DeepSeek-V3.2	64.5	1621.6	72.7	649.9
MATH500	Qwen3.5-9B	98.3	1147.5	97.5	314.6
MATH500	Qwen3.5-35B-A3B	96.7	1076.7	99.2	301.6
MATH500	Kimi-K2.5	97.5	497.6	100.0	155.3
MATH500	GLM-5	99.2	1329.4	95.9	148.8
MATH500	DeepSeek-V3.2	96.7	938.5	99.2	205.5
AIME	Qwen3.5-9B	54.2	3415.9	52.1	869.1
AIME	Qwen3.5-35B-A3B	58.3	3249.2	75.0	1382.3
AIME	Kimi-K2.5	64.6	2002.9	79.2	1148.9
AIME	GLM-5	47.9	3359.9	58.3	1004.1
AIME	DeepSeek-V3.2	54.2	3122.6	68.8	1986.3

The per-listener audit confirms that route choices can be selected without using the held-out half. The route-repair analysis is scientifically important because naive multi-dialect composition can be worse than Raw CoT even when individual dialects are useful. This motivates listener-resistance penalties, route simplicity and the rule-bank emphasis on negative-transfer guards.

A.3 Matched dialect-field ablation

We next examine whether the fields of a dialect card are merely descriptive metadata or whether they affect the route that a listener can execute. To avoid attributing changes to differences in coverage, we report a matched field-ablation analysis: a full MDia route and an ablated route are compared only when both are available for the same benchmark, listener and task identifier. This conservative matching removes some rows but yields a cleaner estimate of field importance.

Table 12 reports the most stable field intervention: removing the explicit symbol inventory from the dialect card. The effect is task-dependent. On GPQA-main, accuracy drops from 72.6% to 59.1%; on MATH500, it drops from 98.7% to 94.1%. In contrast, MMLU-Pro is unchanged, and the matched AIME subset is also unchanged. The aggregate matched accuracy decreases from 85.7% to 80.1%. These results support a narrower claim: the symbol inventory is not decorative on tasks where the listener must preserve scientific alternatives or mathematical state, but its measured effect depends on the task distribution and on the matched coverage of the archive.

The ablation also delimits the claim. MDia does not require every card field to be equally important on every benchmark. Broad multiple-choice knowledge tasks may be solvable from shallow semantic cues, whereas scientific disambiguation and mathematical derivation benefit more from an explicit inventory of compact operators, bindings, and verification tags. The matched analysis therefore supports the mechanism that dialect cards carry executable structure, while leaving stronger causal claims about individual card fields to

Table 12: Matched symbol-inventory ablation. Rows are matched by benchmark, listener and task identifier. Each entry reports accuracy in percent followed by average generated completion tokens. The result is reported as a mechanism analysis rather than as part of the primary benchmark claim.

Benchmark	Matched events	Full MDia	No symbol inventory	Δ acc.
AIME 2024–25	94	58.5 (944)	58.5 (944)	0.0
GPQA-main	430	**72.6** (777)	59.1 (625)	-13.5
MATH500	625	**98.7** (227)	94.1 (242)	-4.6
MMLU-Pro	400	85.8 (201)	85.8 (201)	0.0
All matched	1549	**85.7** (416)	80.1 (380)	-5.6

larger fully matched evaluations.

A.4 Leave-one-family stress test

We additionally remove one dialect family from the routing archive and re-evaluate the held-out route. Table 13 shows that the aggregate result is stable when a single family is removed from this archive. This supports a limited robustness claim: the router is not brittle to a single named family under the present benchmark mixture. It should not be interpreted as proof that every family contributes unique information, because public and inherited dialects can remain available after a named family is removed.

Table 13: Leave-one-family routing stress test. Each row removes one dialect family from the routing archive before held-out evaluation. The result supports route stability under single-family removal, but does not by itself prove that every family contributes unique dialect information.

Removed family	Accuracy	Avg. generated tokens
None	84.0	484
DeepSeek	84.0	484
GLM	84.3	511
Kimi	84.0	484
Qwen-large	84.0	484
Qwen-small	84.0	484

A.5 Supplementary transfer matrices

The following tables preserve the validation-style speaker-listener transfer matrices for the four reasoning benchmarks. Rows denote listener models, and columns denote speaker dialect families. Each entry reports accuracy followed by generated completion tokens.

Table 14: Speaker–listener transfer matrix for MMLU-Pro (9x20-ev10). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	66.17 (598.4)	68.16 (552.9)	59.20 (445.3)	65.17 (567.8)	71.14 (689.1)
Qwen3.5-35B-A3B	72.64 (647.5)	76.62 (562.9)	75.12 (371.3)	74.63 (428.8)	78.11 (976.2)
Kimi-K2.5	75.12 (328.1)	80.60 (277.9)	75.62 (315.2)	75.62 (238.8)	79.60 (401.0)
GLM-5	73.13 (261.0)	76.12 (221.5)	72.64 (205.3)	72.64 (192.3)	75.62 (300.7)
DeepSeek-V3.2	75.62 (343.4)	79.10 (288.4)	69.15 (290.7)	77.11 (315.6)	79.10 (472.7)

Table 15: Speaker–listener transfer matrix for MMLU-Pro (18x10-ev5). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	71.14 (633.2)	64.68 (280.1)	64.68 (364.9)	68.16 (357.2)	70.65 (577.7)
Qwen3.5-35B-A3B	75.62 (587.4)	67.16 (202.3)	75.12 (473.0)	76.12 (417.2)	77.61 (868.0)
Kimi-K2.5	77.11 (314.2)	79.60 (161.4)	80.10 (243.8)	78.11 (258.4)	79.60 (313.5)
GLM-5	76.62 (275.5)	77.61 (127.4)	76.62 (147.7)	73.63 (118.5)	74.63 (176.1)
DeepSeek-V3.2	76.62 (341.2)	77.61 (192.8)	77.11 (280.7)	73.63 (178.6)	79.60 (359.0)

Table 16: Speaker–listener transfer matrix for GPQA-main (9x20-ev10). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	53.57 (2031.4)	29.02 (967.7)	48.66 (1467.4)	52.68 (1470.4)	50.45 (1497.3)
Qwen3.5-35B-A3B	64.73 (1773.4)	31.70 (1109.2)	51.34 (1808.4)	54.46 (1597.0)	55.36 (1970.8)
Kimi-K2.5	66.96 (896.1)	33.04 (811.1)	63.39 (846.8)	59.38 (687.3)	60.27 (984.9)
GLM-5	71.43 (607.1)	47.77 (445.1)	56.25 (489.3)	61.61 (394.7)	59.38 (480.9)
DeepSeek-V3.2	66.07 (947.4)	47.32 (816.9)	57.14 (969.7)	56.70 (938.3)	49.11 (1186.3)

Table 17: Speaker–listener transfer matrix for GPQA-main (18x10-ev5). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	52.68 (1705.6)	50.45 (1817.7)	50.45 (1249.0)	53.57 (1562.2)	56.70 (1763.5)
Qwen3.5-35B-A3B	61.16 (1463.2)	64.29 (1870.8)	58.93 (1160.7)	61.61 (1535.9)	59.38 (1937.0)
Kimi-K2.5	59.82 (726.6)	56.70 (901.4)	60.71 (636.6)	47.77 (668.6)	62.50 (752.9)
GLM-5	55.80 (366.2)	60.27 (659.1)	57.59 (333.6)	60.71 (430.4)	58.48 (679.4)
DeepSeek-V3.2	54.46 (818.8)	64.29 (1140.0)	54.02 (1093.7)	58.93 (958.3)	58.48 (1035.7)

Table 18: Speaker–listener transfer matrix for MATH500 (9x20-ev10). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	82.80 (436.6)	81.20 (459.8)	67.60 (418.6)	70.80 (517.1)	73.60 (408.8)
Qwen3.5-35B-A3B	94.00 (442.5)	94.80 (451.0)	78.00 (367.9)	88.80 (553.3)	77.20 (282.1)
Kimi-K2.5	94.40 (219.3)	90.40 (292.1)	90.40 (287.3)	94.80 (245.9)	91.60 (244.1)
GLM-5	80.80 (193.4)	80.80 (198.9)	76.00 (205.0)	74.80 (204.9)	74.80 (231.7)
DeepSeek-V3.2	90.80 (281.8)	93.60 (327.6)	89.20 (349.6)	84.40 (367.3)	91.60 (403.6)

Table 19: Speaker–listener transfer matrix for MATH500 (18x10-ev5). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	75.60 (492.1)	87.20 (541.5)	78.00 (486.6)	76.80 (442.8)	89.60 (560.8)
Qwen3.5-35B-A3B	92.00 (391.7)	92.00 (546.0)	82.80 (375.3)	92.00 (475.3)	92.80 (594.3)
Kimi-K2.5	93.60 (210.2)	94.40 (263.9)	91.60 (209.4)	95.60 (262.4)	94.40 (380.9)
GLM-5	80.00 (181.3)	83.20 (265.7)	70.40 (97.2)	86.00 (207.0)	82.40 (358.0)
DeepSeek-V3.2	91.60 (265.6)	94.00 (374.5)	88.00 (237.4)	94.40 (337.0)	96.40 (550.8)

Table 20: Speaker–listener transfer matrix for AIME 2024–25 (9x20-ev10). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	38.83 (3324.0)	35.92 (2995.2)	36.89 (2630.7)	34.95 (3187.2)	37.86 (3080.7)
Qwen3.5-35B-A3B	55.34 (3095.3)	49.51 (2882.9)	32.04 (2934.9)	46.60 (2899.9)	50.49 (2909.6)
Kimi-K2.5	66.99 (2045.6)	56.31 (1961.7)	52.43 (1804.5)	57.28 (1841.9)	62.14 (1745.7)
GLM-5	33.01 (2370.9)	30.10 (1907.0)	25.24 (1594.3)	30.10 (1853.9)	27.18 (1615.1)
DeepSeek-V3.2	56.31 (2969.1)	55.34 (2510.9)	47.57 (2613.6)	55.34 (2588.0)	55.34 (2476.4)

Table 21: Speaker–listener transfer matrix for AIME 2024–25 (18x10-ev5). Cells report accuracy (tokens).

Listener \ Speaker	Qwen3.5-9B	Qwen3.5-35B-A3B	Kimi-K2.5	GLM-5	DeepSeek-V3.2
Qwen3.5-9B	22.33 (2562.1)	39.81 (811.9)	35.92 (3130.8)	33.98 (3304.6)	37.86 (3056.5)
Qwen3.5-35B-A3B	42.72 (2178.2)	52.43 (1565.7)	58.25 (3024.0)	52.43 (3076.3)	53.40 (2880.6)
Kimi-K2.5	49.51 (1787.4)	43.69 (1639.4)	53.40 (1862.9)	57.28 (1920.3)	60.19 (1639.5)
GLM-5	22.33 (951.2)	20.39 (699.5)	20.39 (1449.9)	22.33 (1662.4)	28.16 (1524.7)
DeepSeek-V3.2	53.40 (2177.9)	41.75 (1859.6)	51.46 (2573.2)	57.28 (2701.8)	49.51 (2548.1)

A.6 Profile-ordered transfer geometry

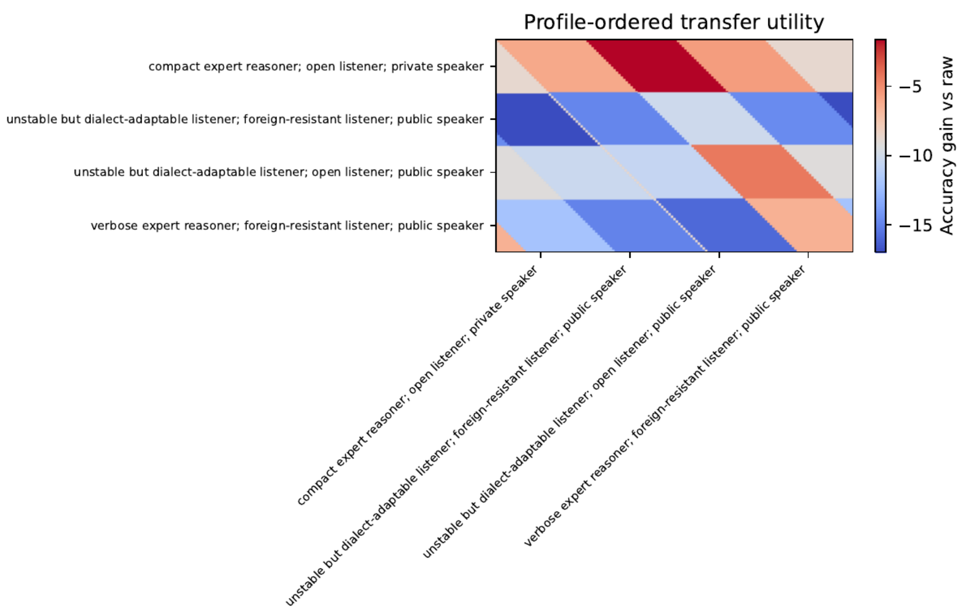

Figure 2: Profile-ordered transfer geometry. The heatmap reorders speaker and listener axes by inferred model profiles, making receiver-relative and task-conditioned transfer structure visible.

A.7 Cross-generational archive influence analysis

The fixed-archive functional-influence analysis is summarized in the Results table and by the survival curves below. It measures how archived dialect families affect downstream route utility; it is not evidence of parameter-level or live online language evolution.

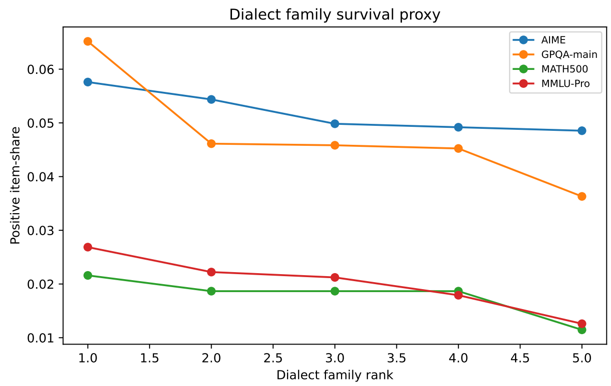

Figure 3: Dialect-family survival proxy. The curves summarize fixed-archive functional survival across benchmark families and support the transfer-structure analysis.

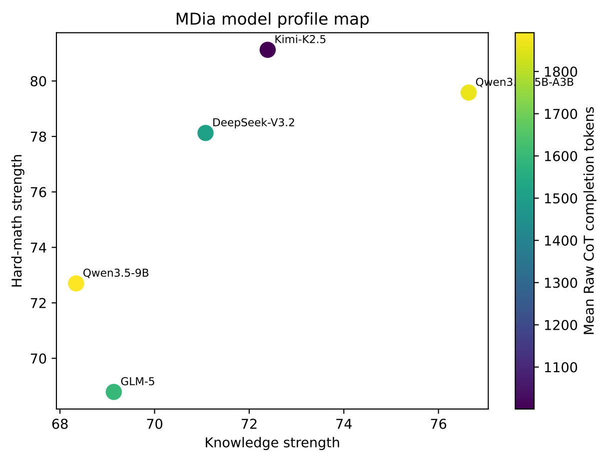

Figure 4: Model-profile map. The map supports profile-level interpretation by positioning models according to raw capability, verbosity and transfer-related attributes.

A.8 Full machine-sociolinguistic rule bank

Table 22: Full 100-rule machine-sociolinguistic bank. Scores summarize empirical support in the MDia rule audit.

ID	Taxonomy	Rule	Score	Level	Statement
R001	Receiver/listener rules	Receiver-relative utility	0.71	strong	A dialect's utility depends on listener and task, not only on speaker quality.
R002	Receiver/listener rules	Listener openness asymmetry	1.00	full	Listeners differ in how much they benefit from foreign dialects.
R003	Receiver/listener rules	Expert resistance	0.34	weak	High-performing listeners may resist simplifying or foreign dialects.
R004	Receiver/listener rules	Compression-fragile listeners	0.81	strong	Some listeners lose accuracy when dialects are too compact.
R005	Receiver/listener rules	Listener variance dominance	0.65	partial	Receiver-side variation can dominate source-speaker variation in some tasks.
R006	Receiver/listener rules	Listener-specific parse fragility	0.35	weak	Parse failures and malformed final answers vary systematically by listener.
R007	Receiver/listener rules	Listener-specific verification demand	0.35	weak	Some listeners need explicit verification tags to retain correctness.
R008	Receiver/listener rules	Listener-specific redundancy preference	0.35	weak	Some listeners benefit from redundant structure while others prefer compact traces.
R009	Receiver/listener rules	Foreign-dialect risk	0.93	full	Foreign dialects can harm listeners even when average transfer is positive.
R010	Receiver/listener rules	Same-family listener accommodation	0.42	boundary	Models from the same family or scale line transfer dialects more easily.
R011	Receiver/listener rules	Cross-family listener friction	0.42	boundary	Cross-family dialect transfer is more fragile than within-family transfer.
R012	Receiver/listener rules	Small-listener scaffold benefit	0.35	weak	Smaller listeners can benefit from public scaffolding dialects.
R013	Receiver/listener rules	Large-listener simplification constraint	0.35	weak	Larger listeners may lose information under over-simplified dialects.
R014	Receiver/listener rules	Listener temperature sensitivity	0.30	boundary	Listener route utility may change under decoding-temperature changes.
R015	Receiver/listener rules	Listener answer-format brittleness	0.35	weak	Some listeners fail more often from answer-contract violations than reasoning errors.
R016	Speaker/publicness rules	Public dialect asymmetry	1.00	full	Some speakers produce more broadly adoptable dialects.
R017	Speaker/publicness rules	Weak-speaker teaching	0.90	full	Lower raw-accuracy speakers can still teach foreign listeners.
R018	Speaker/publicness rules	Strong-speaker private dialects	0.35	weak	Strong speakers can produce private dialects that do not transfer broadly.
R019	Speaker/publicness rules	Speaker variance dominance	0.35	weak	Source-dialect effects can dominate receiver effects in some tasks.
R020	Speaker/publicness rules	Self-talk is not teaching	0.50	partial	A speaker's self dialect need not be its best teaching dialect.
R021	Speaker/publicness rules	Private dialect specialization	0.40	weak	Some dialects are self-specialized and transfer poorly.
R022	Speaker/publicness rules	Public bridge potential	0.20	unsupported	Public bridge dialects can help otherwise resistant speaker-listener pairs.
R023	Speaker/publicness rules	Speaker concision-publicness tradeoff	0.35	weak	More concise speaker dialects can become less public.
R024	Speaker/publicness rules	Speaker verifier-rich teaching	0.35	weak	Speakers that include verification operators can teach robustly.
R025	Speaker/publicness rules	Speaker overcompression harm	0.35	weak	Speaker-side compression can remove information needed by foreign listeners.

ID	Taxonomy	Rule	Score	Level	Statement
R026	Speaker/publicness rules	Speaker family style inheritance	0.35	weak	Speaker family affects dialect notation and transfer style.
R027	Speaker/publicness rules	Speaker task-specialist transfer	0.60	partial	Task-specialist speakers transfer best within their task family.
R028	Speaker/publicness rules	Speaker generalist bridge role	0.35	weak	Generalist speakers can serve as broad bridge dialect sources.
R029	Speaker/publicness rules	Speaker profile beats raw score	0.35	weak	Speaker profile variables predict transfer better than raw benchmark score alone.
R030	Speaker/publicness rules	Speaker dialect diversity advantage	0.35	weak	Speakers with diverse dialect pools provide better routing candidates.
R031	Task-conditioned rules	Task-conditioned transfer	0.67	partial	Speaker-listener gains differ across benchmark families.
R032	Task-conditioned rules	Hard-math verifier pressure	0.51	partial	Hard math tasks benefit from verifier-rich or longer dialects.
R033	Task-conditioned rules	Knowledge-QA redundancy pressure	0.35	weak	Knowledge QA tasks need redundancy around factual options and answer contracts.
R034	Task-conditioned rules	Code edge-case dialect pressure	0.25	boundary	Code tasks require dialect fields for I/O contracts, edge cases, and complexity.
R035	Task-conditioned rules	Tool-call schema-binding pressure	0.78	strong	Function-calling tasks favor explicit function-selection and argument-binding dialects.
R036	Task-conditioned rules	RAG evidence-chain pressure	0.20	boundary	Evidence reasoning tasks require source IDs and relation-chain operators.
R037	Task-conditioned rules	Narrative entity-state pressure	0.25	boundary	Narrative tasks benefit from entity-state and evidence-tracking dialects.
R038	Task-conditioned rules	Short-answer ceiling effect	0.62	partial	Near-saturated short-answer tasks show smaller accuracy gains.
R039	Task-conditioned rules	Benchmark headroom effect	0.74	strong	MDia gains are larger where baselines leave enough headroom.
R040	Task-conditioned rules	Long-context compression fragility	0.35	weak	Long-context tasks can fail if compression removes grounding details.
R041	Task-conditioned rules	Multi-hop relation tagging benefit	0.35	weak	Multi-hop tasks benefit from explicit relation tags.
R042	Task-conditioned rules	Symbolic transformation operator benefit	0.35	weak	Symbolic tasks benefit from typed operators and transformations.
R043	Task-conditioned rules	Commonsense narrative softness	0.25	boundary	Commonsense narratives need softer, less over-symbolized dialects.
R044	Task-conditioned rules	Retrieval noise amplification	0.20	boundary	Retrieved evidence noise can amplify dialect mismatch.
R045	Task-conditioned rules	Domain-specific dialect specialization	0.35	weak	Domain-specialized dialects transfer best within their domain.
R046	Compression/redundancy rules	Cost-sensitive utility	0.90	full	Token-efficient dialects can move the accuracy-token frontier.
R047	Compression/redundancy rules	Conciseness is not sufficient	0.21	unsupported	Lower token count alone does not predict correctness.
R048	Compression/redundancy rules	Overcompression failure	0.11	unsupported	Large token compression can reduce accuracy.
R049	Compression/redundancy rules	Redundancy as listener insurance	0.35	weak	A little structured redundancy can protect foreign listeners.
R050	Compression/redundancy rules	Verification tags can be worth their tokens	0.35	weak	Explicit check tags can improve net utility despite extra tokens.
R051	Compression/redundancy rules	Symbol table reuse amortizes cost	0.35	weak	Reusable symbol tables become cost-effective across repeated tasks.
R052	Compression/redundancy rules	Short operators survive frequent use	0.35	weak	Short stable operators work best when used repeatedly.

ID	Taxonomy	Rule	Score	Level	Statement
R053	Compression/redundancy rules	Ambiguous abbreviations fail across listeners	0.35	weak	Unclear abbreviations increase cross-listener parse failures.
R054	Compression/redundancy rules	Structured redundancy beats natural verbosity	0.35	weak	Structured redundancy is more efficient than verbose natural language.
R055	Compression/redundancy rules	Token savings saturate before accuracy	0.35	weak	Beyond a threshold, extra compression no longer improves utility.
R056	Compression/redundancy rules	Parser-compatible compactness	0.78	strong	Compactness is useful only if final answers remain parse-compatible.
R057	Compression/redundancy rules	Answer-contract compression	0.78	strong	Answer contracts can be compressed safely when schema is explicit.
R058	Compression/redundancy rules	Dialect entropy vs interpretability	0.35	weak	Higher symbol entropy can reduce foreign-listener interpretability.
R059	Compression/redundancy rules	Compression helps easy tasks more	0.62	partial	Easy tasks tolerate stronger compression.
R060	Compression/redundancy rules	Compression hurts hard tasks without verifier tags	0.35	weak	Hard tasks need verification to survive compression.
R061	Routing/code-switching rules	Pragmatic code-switching	0.50	partial	Different task types favor different dialect sources.
R062	Routing/code-switching rules	Archive routing advantage	0.40	weak	History-aware selection beats random or self-only dialect selection.
R063	Routing/code-switching rules	Profile routing beats model-name routing	0.55	partial	Profile features outperform literal model-name routing.
R064	Routing/code-switching rules	Task-aware routing beats global routing	0.75	strong	Task-conditioned routing beats a single global dialect choice.
R065	Routing/code-switching rules	Route simplicity can beat over-composition	1.00	full	Simple profile-aware routing can beat over-composed multi-dialect routing.
R066	Routing/code-switching rules	Bridge routing helps resistant pairs	0.35	weak	Bridge dialects help when direct transfer is risky.
R067	Routing/code-switching rules	Composition helps hard tasks but not easy ones	0.35	weak	Multi-dialect composition is useful mainly for hard tasks.
R068	Routing/code-switching rules	Random foreign routing is unsafe	0.75	strong	Randomly selected foreign dialects can cause negative transfer.
R069	Routing/code-switching rules	Oracle gap reveals unused social information	0.75	strong	The archive contains unrealized gains beyond the implemented router.
R070	Routing/code-switching rules	Route abstention protects against dialect mismatch	0.52	partial	Routing should abstain when mismatch risk is high.
R071	Routing/code-switching rules	Route diversity improves ensemble robustness	0.35	weak	Diverse route candidates reduce brittleness.
R072	Routing/code-switching rules	Majority vote is weaker than profile-aware vote	0.35	weak	Profile-aware routing beats unweighted votes over dialects.
R073	Routing/code-switching rules	Cost-aware route selection changes winners	0.75	strong	Token-aware scoring changes which dialects are selected.
R074	Routing/code-switching rules	Multi-round routing needs stopping discipline	0.35	weak	Extra routing rounds need stopping criteria to avoid cost blow-up.
R075	Routing/code-switching rules	Rule-aware routing improves stability	0.52	partial	Validated rules can constrain routes and reduce failures.
R076	Archive/evolution rules	Dialect survival is task-specific	0.53	partial	Functional survival varies by benchmark.
R077	Archive/evolution rules	Founder-effect proxy	0.57	partial	Removing source families disproportionately changes pooled outcomes.
R078	Archive/evolution rules	Borrowing proxy	0.23	unsupported	Functional influence is associated with token-structure shifts.
R079	Archive/evolution rules	Cross-generational influence is directed	0.35	weak	Source influence is directed and task-dependent.
R080	Archive/evolution rules	Generation depth improves then saturates	0.38	boundary	More evolution generations help until validation utility saturates.

ID	Taxonomy	Rule	Score	Level	Statement
R081	Archive/evolution rules	Archive diversity prevents early collapse	0.55	partial	Diverse archives prevent premature convergence to brittle dialects.
R082	Archive/evolution rules	Public dialects survive longer	0.55	partial	Public dialects persist under routing better than private dialects.
R083	Archive/evolution rules	Private dialects survive under self-heavy routing	0.35	weak	Self-heavy policies retain private dialects.
R084	Archive/evolution rules	Listener feedback reshapes dialect cards	0.35	weak	Listener outcomes should update dialect metadata.
R085	Archive/evolution rules	Benchmark mixing induces bridge dialects	0.35	weak	Mixed benchmark evolution can induce more public bridge dialects.
R086	Archive/evolution rules	Source removal reveals hidden dependency	0.55	partial	Leave-one-source analysis reveals hidden archive dependencies.
R087	Archive/evolution rules	Repeated transmission increases structure	0.38	boundary	Repeated evolution/transmission increases dialect regularity.
R088	Archive/evolution rules	Mutation repairs ambiguity more than it invents from scratch	0.38	boundary	Mutation mainly repairs ambiguous conventions.
R089	Archive/evolution rules	Archive age affects utility	0.30	boundary	Older dialects can become stale under model/API drift.
R090	Archive/evolution rules	Validation profile drift predicts test risk	0.35	weak	Validation/test profile drift predicts route failure.
R091	Robustness, scale, and deployment rules	Scale does not monotonically increase publicness	0.42	boundary	Scale does not guarantee public dialect utility.
R092	Robustness, scale, and deployment rules	Scale mismatch creates dialect mismatch	0.42	boundary	Large scale gaps can make dialects harder to transfer.
R093	Robustness, scale, and deployment rules	Same-family scale transfer is easier than cross-family transfer	0.42	boundary	Same-family scale transfer is easier than cross-family transfer.
R094	Robustness, scale, and deployment rules	API model drift changes social profiles	0.30	boundary	Hosted model updates can change social/dialect profiles.
R095	Robustness, scale, and deployment rules	Cache-aware cost can change route preference	0.35	weak	Cache-aware accounting can change the selected route.
R096	Robustness, scale, and deployment rules	Seed-stable rules are stronger than single-run rules	0.58	partial	Rules replicated across seeds are stronger than single-run patterns.
R097	Robustness, scale, and deployment rules	Parse-failure diagnostics predict route failure	0.35	weak	Parse failure rates predict bad routes.
R098	Robustness, scale, and deployment rules	Human-readable finalization is needed for auditability	0.72	strong	Final answers need readable contracts for audit and paper evaluation.
R099	Robustness, scale, and deployment rules	Tool/RAG dialects require stricter answer contracts	0.78	strong	Tool/RAG dialects need stricter final schemas than short QA.
R100	Robustness, scale, and deployment rules	Negative-transfer guards are essential for safe deployment	0.75	strong	Routes need guards against dialect mismatch and negative transfer.

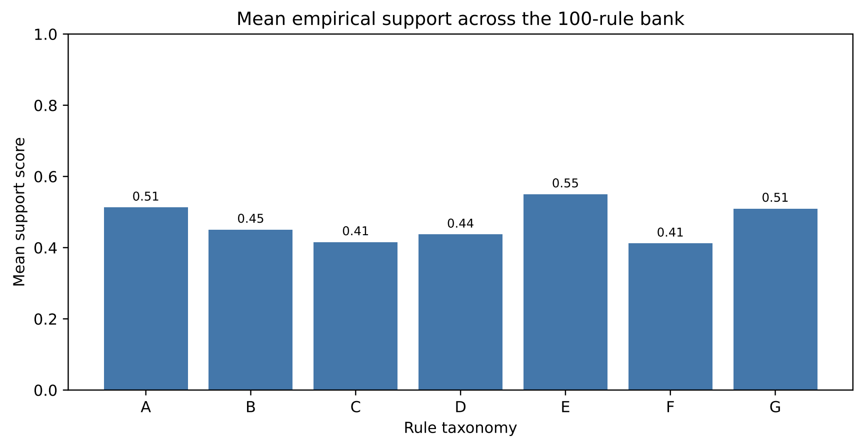

Figure 5: Rule-bank taxonomy. The 100 rules are grouped by receiver, speaker, task, compression, routing, archive, robustness, interpretability and deployment functions.

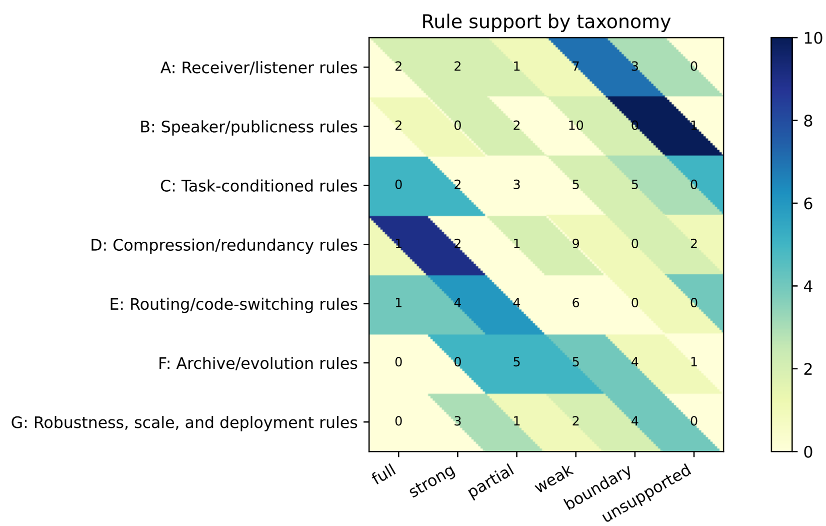

Figure 6: Rule-support heatmap. Full and strong rules are concentrated in listener openness, publicness, route simplicity, foreign-dialect risk, weak-speaker teaching and cost-sensitive utility. Boundary and unsupported rules are retained as constraints on overclaiming.

The main parts use the supported subset of rules to explain MDia's empirical behavior. The complete table is retained here so that the rule bank remains falsifiable and extensible. Rules with limited or boundary-case support are not used as main performance claims. They are useful boundary cases for deciding which sociolinguistic analogies are too broad, which metrics are too coarse, and which claims require cleaner held-out evaluation.

A.9 Rule-by-rule analyses for the 100-rule bank

The following entries expand the 100-rule bank beyond the compact tabular summary. Each entry states the machine-sociolinguistic meaning of the rule, the experimental feature used to audit it, the observed support level, and the routing implication for MDia. The support labels should be read as empirical strengths within the present MDia archive, not as universal laws about all model families or all tasks.

R001: Receiver-relative utility.

Meaning. A dialect's utility depends on listener and task, not only on speaker quality.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.71, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R002: Listener openness asymmetry.

Meaning. Listeners differ in how much they benefit from foreign dialects.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 1.00, classified as full. The support level is full: within the audited evidence stream, the predicted pattern is consistently present and can serve as a primary routing prior.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia can assign this feature high weight when selecting or filtering dialect routes.

R003: Expert resistance.

Meaning. High-performing listeners may resist simplifying or foreign dialects.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.34, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R004: Compression-fragile listeners.

Meaning. Some listeners lose accuracy when dialects are too compact.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.81, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R005: Listener variance dominance.

Meaning. Receiver-side variation can dominate source-speaker variation in some tasks.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.65, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R006: Listener-specific parse fragility.

Meaning. Parse failures and malformed final answers vary systematically by listener.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R007: Listener-specific verification demand.

Meaning. Some listeners need explicit verification tags to retain correctness.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R008: Listener-specific redundancy preference.

Meaning. Some listeners benefit from redundant structure while others prefer compact traces.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R009: Foreign-dialect risk.

Meaning. Foreign dialects can harm listeners even when average transfer is positive.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.93, classified as full. The support level is full: within the audited evidence stream, the predicted pattern is consistently present and can serve as a primary routing prior.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia can assign this feature high weight when selecting or filtering dialect routes.

R010: Same-family listener accommodation.

Meaning. Models from the same family or scale line transfer dialects more easily.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.42, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should use this feature as a guardrail or boundary detector.

R011: Cross-family listener friction.

Meaning. Cross-family dialect transfer is more fragile than within-family transfer.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.42, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should use this feature as a guardrail or boundary detector.

R012: Small-listener scaffold benefit.

Meaning. Smaller listeners can benefit from public scaffolding dialects.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R013: Large-listener simplification constraint.

Meaning. Larger listeners may lose information under over-simplified dialects.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R014: Listener temperature sensitivity.

Meaning. Listener route utility may change under decoding-temperature changes.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.30, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should use this feature as a guardrail or boundary detector.

R015: Listener answer-format brittleness.

Meaning. Some listeners fail more often from answer-contract violations than reasoning errors.

Experimental feature. The audit compares each listener's raw, self-dialect, and foreign-dialect outcomes across benchmarks, then measures how much route utility is explained by listener identity, openness, parse robustness, and compression tolerance.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Condition route scores on listener profile, parse risk, and compression tolerance instead of assuming that one dialect is universally interpretable. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R016: Public dialect asymmetry.

Meaning. Some speakers produce more broadly adoptable dialects.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 1.00, classified as full. The support level is full: within the audited evidence stream, the predicted pattern is consistently present and can serve as a primary routing prior.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia can assign this feature high weight when selecting or filtering dialect routes.

R017: Weak-speaker teaching.

Meaning. Lower raw-accuracy speakers can still teach foreign listeners.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.90, classified as full. The support level is full: within the audited evidence stream, the predicted pattern is consistently present and can serve as a primary routing prior.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia can assign this feature high weight when selecting or filtering dialect routes.

R018: Strong-speaker private dialects.

Meaning. Strong speakers can produce private dialects that do not transfer broadly.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R019: Speaker variance dominance.

Meaning. Source-dialect effects can dominate receiver effects in some tasks.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R020: Self-talk is not teaching.

Meaning. A speaker's self dialect need not be its best teaching dialect.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.50, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R021: Private dialect specialization.

Meaning. Some dialects are self-specialized and transfer poorly.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.40, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R022: Public bridge potential.

Meaning. Public bridge dialects can help otherwise resistant speaker-listener pairs.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.20, classified as unsupported. The support level is unsupported: the audit does not justify using this rule as a positive prior, so it is retained mainly as a control or cautionary case.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should not reward this feature unless future archive evidence changes its status.

R023: Speaker concision-publicness tradeoff.

Meaning. More concise speaker dialects can become less public.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R024: Speaker verifier-rich teaching.

Meaning. Speakers that include verification operators can teach robustly.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R025: Speaker overcompression harm.

Meaning. Speaker-side compression can remove information needed by foreign listeners.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R026: Speaker family style inheritance.

Meaning. Speaker family affects dialect notation and transfer style.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R027: Speaker task-specialist transfer.

Meaning. Task-specialist speakers transfer best within their task family.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.60, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R028: Speaker generalist bridge role.

Meaning. Generalist speakers can serve as broad bridge dialect sources.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R029: Speaker profile beats raw score.

Meaning. Speaker profile variables predict transfer better than raw benchmark score alone.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R030: Speaker dialect diversity advantage.

Meaning. Speakers with diverse dialect pools provide better routing candidates.

Experimental feature. The audit aggregates each speaker dialect over foreign listeners and contrasts publicness with speaker raw accuracy, self-talk behavior, transfer variance, and task affinity.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Select speaker dialects by measured publicness and task affinity, not by raw model strength alone. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R031: Task-conditioned transfer.

Meaning. Speaker-listener gains differ across benchmark families.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.67, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R032: Hard-math verifier pressure.

Meaning. Hard math tasks benefit from verifier-rich or longer dialects.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.51, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R033: Knowledge-QA redundancy pressure.

Meaning. Knowledge QA tasks need redundancy around factual options and answer contracts.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R034: Code edge-case dialect pressure.

Meaning. Code tasks require dialect fields for I/O contracts, edge cases, and complexity.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.25, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a guardrail or boundary detector.

R035: Tool-call schema-binding pressure.

Meaning. Function-calling tasks favor explicit function-selection and argument-binding dialects.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.78, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R036: RAG evidence-chain pressure.

Meaning. Evidence reasoning tasks require source IDs and relation-chain operators.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.20, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a guardrail or boundary detector.

R037: Narrative entity-state pressure.

Meaning. Narrative tasks benefit from entity-state and evidence-tracking dialects.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.25, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a guardrail or boundary detector.

R038: Short-answer ceiling effect.

Meaning. Near-saturated short-answer tasks show smaller accuracy gains.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.62, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R039: Benchmark headroom effect.

Meaning. MDia gains are larger where baselines leave enough headroom.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.74, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R040: Long-context compression fragility.

Meaning. Long-context tasks can fail if compression removes grounding details.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R041: Multi-hop relation tagging benefit.

Meaning. Multi-hop tasks benefit from explicit relation tags.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R042: Symbolic transformation operator benefit.

Meaning. Symbolic tasks benefit from typed operators and transformations.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R043: Commonsense narrative softness.

Meaning. Commonsense narratives need softer, less over-symbolized dialects.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.25, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a guardrail or boundary detector.

R044: Retrieval noise amplification.

Meaning. Retrieved evidence noise can amplify dialect mismatch.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.20, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should use this feature as a guardrail or boundary detector.

R045: Domain-specific dialect specialization.

Meaning. Domain-specialized dialects transfer best within their domain.

Experimental feature. The audit stratifies route utility by benchmark family and answer format, distinguishing hard-math, multiple-choice, short-answer, schema-binding, evidence-chain, and narrative settings.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use benchmark and answer-format tags as first-class routing features, and avoid transferring a dialect outside the task regime where it is supported. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R046: Cost-sensitive utility.

Meaning. Token-efficient dialects can move the accuracy-token frontier.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.90, classified as full. The support level is full: within the audited evidence stream, the predicted pattern is consistently present and can serve as a primary routing prior.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia can assign this feature high weight when selecting or filtering dialect routes.

R047: Conciseness is not sufficient.

Meaning. Lower token count alone does not predict correctness.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.21, classified as unsupported. The support level is unsupported: the audit does not justify using this rule as a positive prior, so it is retained mainly as a control or cautionary case.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should not reward this feature unless future archive evidence changes its status.

R048: Overcompression failure.

Meaning. Large token compression can reduce accuracy.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.11, classified as unsupported. The support level is unsupported: the audit does not justify using this rule as a positive prior, so it is retained mainly as a control or cautionary case.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should not reward this feature unless future archive evidence changes its status.

R049: Redundancy as listener insurance.

Meaning. A little structured redundancy can protect foreign listeners.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R050: Verification tags can be worth their tokens.

Meaning. Explicit check tags can improve net utility despite extra tokens.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R051: Symbol table reuse amortizes cost.

Meaning. Reusable symbol tables become cost-effective across repeated tasks.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R052: Short operators survive frequent use.

Meaning. Short stable operators work best when used repeatedly.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R053: Ambiguous abbreviations fail across listeners.

Meaning. Unclear abbreviations increase cross-listener parse failures.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R054: Structured redundancy beats natural verbosity.

Meaning. Structured redundancy is more efficient than verbose natural language.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R055: Token savings saturate before accuracy.

Meaning. Beyond a threshold, extra compression no longer improves utility.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R056: Parser-compatible compactness.

Meaning. Compactness is useful only if final answers remain parse-compatible.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.78, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R057: Answer-contract compression.

Meaning. Answer contracts can be compressed safely when schema is explicit.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.78, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R058: Dialect entropy vs interpretability.

Meaning. Higher symbol entropy can reduce foreign-listener interpretability.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R059: Compression helps easy tasks more.

Meaning. Easy tasks tolerate stronger compression.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.62, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R060: Compression hurts hard tasks without verifier tags.

Meaning. Hard tasks need verification to survive compression.

Experimental feature. The audit links the rule to route utility, correctness, token cost, and speaker–listener metadata in the MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Use the rule as a measured route feature rather than as a model-name anecdote. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R061: Pragmatic code-switching.

Meaning. Different task types favor different dialect sources.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.50, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R062: Archive routing advantage.

Meaning. History-aware selection beats random or self-only dialect selection.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.40, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R063: Profile routing beats model-name routing.

Meaning. Profile features outperform literal model-name routing.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.55, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R064: Task-aware routing beats global routing.

Meaning. Task-conditioned routing beats a single global dialect choice.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.75, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R065: Route simplicity can beat over-composition.

Meaning. Simple profile-aware routing can beat over-composed multi-dialect routing.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 1.00, classified as full. The support level is full: within the audited evidence stream, the predicted pattern is consistently present and can serve as a primary routing prior.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia can assign this feature high weight when selecting or filtering dialect routes.

R066: Bridge routing helps resistant pairs.

Meaning. Bridge dialects help when direct transfer is risky.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R067: Composition helps hard tasks but not easy ones.

Meaning. Multi-dialect composition is useful mainly for hard tasks.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R068: Random foreign routing is unsafe.

Meaning. Randomly selected foreign dialects can cause negative transfer.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.75, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R069: Oracle gap reveals unused social information.

Meaning. The archive contains unrealized gains beyond the implemented router.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.75, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R070: Route abstention protects against dialect mismatch.

Meaning. Routing should abstain when mismatch risk is high.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.52, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R071: Route diversity improves ensemble robustness.

Meaning. Diverse route candidates reduce brittleness.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R072: Majority vote is weaker than profile-aware vote.

Meaning. Profile-aware routing beats unweighted votes over dialects.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R073: Cost-aware route selection changes winners.

Meaning. Token-aware scoring changes which dialects are selected.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.75, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R074: Multi-round routing needs stopping discipline.

Meaning. Extra routing rounds need stopping criteria to avoid cost blow-up.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R075: Rule-aware routing improves stability.

Meaning. Validated rules can constrain routes and reduce failures.

Experimental feature. The audit compares self-dialect, foreign-dialect, multi-dialect, guarded, and cost-aware route policies under the same accuracy-token utility.

Result analysis. The normalized support score is 0.52, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Prefer guarded route selection with abstention or fallback when the archive predicts foreign-dialect risk. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R076: Dialect survival is task-specific.

Meaning. Functional survival varies by benchmark.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.53, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R077: Founder-effect proxy.

Meaning. Removing source families disproportionately changes pooled outcomes.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.57, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R078: Borrowing proxy.

Meaning. Functional influence is associated with token-structure shifts.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.23, classified as unsupported. The support level is unsupported: the audit does not justify using this rule as a positive prior, so it is retained mainly as a control or cautionary case.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should not reward this feature unless future archive evidence changes its status.

R079: Cross-generational influence is directed.

Meaning. Source influence is directed and task-dependent.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R080: Generation depth improves then saturates.

Meaning. More evolution generations help until validation utility saturates.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.38, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a guardrail or boundary detector.

R081: Archive diversity prevents early collapse.

Meaning. Diverse archives prevent premature convergence to brittle dialects.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.55, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R082: Public dialects survive longer.

Meaning. Public dialects persist under routing better than private dialects.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.55, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R083: Private dialects survive under self-heavy routing.

Meaning. Self-heavy policies retain private dialects.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R084: Listener feedback reshapes dialect cards.

Meaning. Listener outcomes should update dialect metadata.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R085: Benchmark mixing induces bridge dialects.

Meaning. Mixed benchmark evolution can induce more public bridge dialects.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R086: Source removal reveals hidden dependency.

Meaning. Leave-one-source analysis reveals hidden archive dependencies.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.55, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R087: Repeated transmission increases structure.

Meaning. Repeated evolution/transmission increases dialect regularity.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.38, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a guardrail or boundary detector.

R088: Mutation repairs ambiguity more than it invents from scratch.

Meaning. Mutation mainly repairs ambiguous conventions.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.38, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a guardrail or boundary detector.

R089: Archive age affects utility.

Meaning. Older dialects can become stale under model/API drift.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.30, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should use this feature as a guardrail or boundary detector.

R090: Validation profile drift predicts test risk.

Meaning. Validation/test profile drift predicts route failure.

Experimental feature. The audit studies dialect survival, source removal, transfer reuse, metadata update, and cross-source influence in the saved MDia archive.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Maintain archive metadata for source influence, survival, and reuse so that routing policies can update without relying on model-name anecdotes. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R091: Scale does not monotonically increase publicness.

Meaning. Scale does not guarantee public dialect utility.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.42, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should use this feature as a guardrail or boundary detector.

R092: Scale mismatch creates dialect mismatch.

Meaning. Large scale gaps can make dialects harder to transfer.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.42, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should use this feature as a guardrail or boundary detector.

R093: Same-family scale transfer is easier than cross-family transfer.

Meaning. Same-family scale transfer is easier than cross-family transfer.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.42, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should use this feature as a guardrail or boundary detector.

R094: API model drift changes social profiles.

Meaning. Hosted model updates can change social/dialect profiles.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.30, classified as boundary. The support level is boundary: the rule is most useful for identifying when a dialect assumption may fail, rather than for asserting a universal transfer pattern.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should use this feature as a guardrail or boundary detector.

R095: Cache-aware cost can change route preference.

Meaning. Cache-aware accounting can change the selected route.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R096: Seed-stable rules are stronger than single-run rules.

Meaning. Rules replicated across seeds are stronger than single-run patterns.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.58, classified as partial. The support level is partial: the direction is informative, but exceptions across tasks or listeners make the rule a soft prior rather than a deterministic law.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should use this feature as a soft score adjustment rather than a hard routing constraint.

R097: Parse-failure diagnostics predict route failure.

Meaning. Parse failure rates predict bad routes.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.35, classified as weak. The support level is weak: the pattern is retained as a diagnostic hypothesis and should receive low weight until additional evidence strengthens it.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia should log this feature and use it only as a low-weight diagnostic signal.

R098: Human-readable finalization is needed for auditability.

Meaning. Final answers need readable contracts for audit and paper evaluation.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.72, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R099: Tool/RAG dialects require stricter answer contracts.

Meaning. Tool/RAG dialects need stricter final schemas than short QA.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.78, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

R100: Negative-transfer guards are essential for safe deployment.

Meaning. Routes need guards against dialect mismatch and negative transfer.

Experimental feature. The audit treats seed stability, model-version continuity, parser robustness, scale-family transfer, and negative-transfer screening as deployment variables.

Result analysis. The normalized support score is 0.75, classified as strong. The support level is strong: the pattern is stable enough to support a mechanism claim, while remaining conditional on the evaluated benchmarks and listeners.

Routing implication. Gate deployment on stability, parser validity, and negative-transfer checks before increasing reliance on foreign dialects. Operationally, MDia can assign this feature substantial weight, with task-conditioned checks.

A.10 Controlled schema-binding analysis

This controlled analysis assesses whether dialect cards can impose stricter answer contracts in a tool-use setting. It is separate from the main comparison of benchmarks in Table 1. The result is used as mechanism evidence for rules about schema binding, parse compatibility, and answer-contract compression.

A.11 Implementation and reproducibility notes

All experiments are black-box or API-style inference experiments on frozen models. The local experimental environment uses 8 NVIDIA RTX 4090 GPUs for smaller-model inference and API calls for larger hosted

Table 23: Controlled schema-binding analysis. The analysis uses BFCL-style controlled tasks rather than the official BFCL benchmark and is reported as a secondary probe of answer-contract reliability.

Model	Method/strategy	Success	Parse rate / rows	Completion tokens
Pro/zai-org/GLM-5	MDia schema binding	71.43%	100.00%	104.1
Pro/zai-org/GLM-5	MDia verified	71.43%	100.00%	118.5
Pro/zai-org/GLM-5	Raw JSON	64.29%	100.00%	39.0
Qwen/Qwen3.5-9B	MDia schema binding	57.14%	100.00%	67.3
Qwen/Qwen3.5-9B	MDia verified	50.00%	100.00%	136.9
Qwen/Qwen3.5-9B	Raw JSON	42.86%	100.00%	47.2
Route summary	MDia route	64.29%	14	52.2
Route summary	MDia schema binding	64.29%	14	85.9
Route summary	MDia verified	57.14%	14	127.6
Route summary	Raw JSON	57.14%	14	44.4

models. The key manuscript tables are exported from a normalized JSONL and Parquet archive with fields for query identifiers, speakers, listeners, dialect identifiers, metadata route keys, route plans, parsed answers, correctness, token counts, latency, cost, and source paths. The project distinguishes three empirical streams: held-out main-table results, split-validation route-selection validation results, and validation-style mechanism analyzes. Only the first stream is used for the primary accuracy-token table. The second stream supports the dynamic archive claim. The third stream supports machine-sociolinguistic interpretation.

A.12 Dynamic archive and routing variants

Table 24: Dynamic MDia route-policy components. These components share the same archive but use different retrieval, scoring or composition rules.

Component	Routing mechanism	Scientific role	Reporting status
Archive retrieval	Retrieve dialect records by benchmark family, query profile, listener profile and token budget.	Tests whether the archive can function as reusable dialect memory.	Supported by route-selection diagnostics.
Profile scoring	Weight candidate dialects by publicness, openness, resistance, verbosity and task strengths.	Converts model-name anecdotes into profile-level claims.	Used in Results interpretation.
Dialect composition	Combine planner, solver and verifier dialects when a query appears to require multi-step reasoning.	Explains selective extra-token allocation in hard tasks.	Selective evidence only.
Bridge routing	Insert a public dialect between a private speaker and a resistant listener.	Formalizes accommodation-like transfer.	Design-space diagnostic.
Rule-aware scoring	Penalize route choices that violate supported sociolinguistic rules.	Links rule validation to inference.	Supplementary diagnostic.
Cost-aware scoring	Maximize expected correctness under token and API-cost constraints.	Deployment-oriented extension.	Cost-sensitive analysis.

The route-policy components in Supplementary Table 24 are operational views of the same MDia archive, not separate methods or internal version labels. Single-dialect selection is the most conservative policy: it chooses one dialect family according to validation utility for the target listener and benchmark. Archive retrieval generalizes this policy by searching nearest archived records using benchmark tags, query type, answer format, listener profile and budget. Profile scoring replaces literal model names with behavioral descriptors, which makes the policy more robust when a commercial model is replaced by another model with similar behavior. Bridge routing is used when direct transfer from a specialist speaker to a resistant listener is unreliable. Dialect composition invokes planner, solver and verifier dialects only when a difficulty or uncertainty estimate predicts that a single dialect may be insufficient. Rule-aware and cost-aware scoring apply the validated rule bank, generated-token budgets, latency and monetary cost as constraints on routing. Bandit-style updating can turn the archive into an online learner after feedback labels become available.

A practical MDia deployment can be described as a repeated archive-update loop. Given a new query x, the system first computes a lightweight query profile $\phi(x)$. This profile includes benchmark-like tags, answer format, domain, estimated difficulty, need for calculation, need for option discrimination and allowable token budget. The system then retrieves archived records whose query profiles are close to $\phi(x)$ and whose listener profile matches the target model. Candidate dialects are scored by expected utility,

$$S(D_j, M_i, x) = \widehat{A}_{i\leftarrow j}(x) - \lambda_{\text{tok}}\widehat{C}_{i\leftarrow j}(x) - \lambda_{\text{risk}}\widehat{R}_{i\leftarrow j}(x) + \lambda_{\text{pub}}\widehat{P}_j, \tag{19}$$

where $\widehat{A}$ is expected correctness, $\widehat{C}$ is expected generated-token cost, $\widehat{R}$ is estimated foreign-dialect risk and $\widehat{P}$ is speaker publicness. The policy selects the highest-scoring single dialect when the margin is large. It selects a bridge or verifier composition when the predicted risk is high or the difficulty estimate exceeds a threshold. After execution, the response, parsed answer, correctness when available, tokens, latency and cost are appended to the archive. This makes MDia a maintainable workflow rather than a fixed prompt list.

The archive also supports conservative deployment. A production system can default to Raw CoT or a verified self-dialect when the archive is sparse. It can enable foreign dialects only when listener openness has been estimated on validation records. It can require verifier dialects for AIME-like problems and forbid ultra-compact dialects for compression-fragile listeners. These policies follow directly from the supported rules in Table 5 and the full rule bank in Table 22.

A.13 Benchmark-specific dialect demands

MMLU-Pro and GPQA-main both require knowledge discrimination, but they stress different dialectal capacities. MMLU-Pro benefits from compact option tracking, domain tags, elimination operators and concise evidence summaries. A useful dialect for this setting often has symbols for candidate answer states, contraindications, confidence changes and final answer emission. The dialect does not need to expose a long explanation; it needs to preserve enough state for the listener to distinguish plausible options. This explains why MDia obtains large MMLU-Pro gains while reducing tokens substantially.

GPQA-main is more adversarial and expert-oriented. The useful dialects are less like simple abbreviation systems and more like evidence-preserving schemas. They need to mark assumptions, scientific relations, tensions and weakly supported options. This benchmark is where listener openness and public dialects become especially important. A compact dialect that works for one listener may not preserve enough specialist evidence for another listener. The router therefore benefits from profile-aware route selection rather than unconditional prompt compression.

MATH500 is close to saturation for several raw listeners. The main contribution of MDia on this benchmark is therefore not a large absolute accuracy jump in every condition. Instead, MDia shows that high accuracy can be maintained with far fewer generated tokens. The useful dialects preserve variable bindings, equation transformations, subgoal labels and short verification tags. They remove narrative explanation but keep the algebraic state that the listener needs. This is why the benchmark-level token reduction is large even when the accuracy margin is small.

AIME is the clearest hard-math stress test. Raw CoT spends thousands of tokens because the problems require search, multi-step derivation and careful final checking. AIME dialects that survive in the archive usually contain verifier-rich operators: constraint checks, modular arithmetic markers, case labels, boundary tests and final answer validation. The cross-generational functional influence analysis shows especially large source-family effects on AIME, which is consistent with strong selection pressure for reusable hard-math strategies. This is also why a dynamic router should not always choose the shortest dialect for AIME-like queries. It should choose a dialect whose compactness preserves verification rather than deleting it.

BFCL-v3 stresses a different kind of symbolic fragility: the final answer is not an explanation but a parseable function-call object. A useful dialect therefore preserves function names, argument slots, default values, nested structures and the ordering of parallel calls. The route should not reward a trace that is semantically plausible but fails the evaluator because one key is omitted or a multi-call request is collapsed into one call. This benchmark makes the answer contract itself part of the dialect.

LiveCodeBench-output asks the model to predict exact program output. The compact dialect must preserve execution state, branch conditions, loop counters, mutation points and stdout normalization. Unlike MATH500, the final parse may depend on whitespace, line order or a silent state update rather than on a symbolic mathematical expression. This is why a schema-first stdout dialect can be useful: it allows the listener to carry state internally while emitting only the normalized output.

MultiHopRAG requires evidence-grounded hops across supplied documents. Useful dialects preserve entity bridges, support identifiers, contradiction tags and a null-support guard. The null-query condition is particularly important because an overcompressed dialect can turn missing evidence into an unsupported guess. For this benchmark, the router should select a direct evidence dialect for comparison questions, a verifier dialect for inference questions and a null-aware dialect when support may be absent.

MuSR stresses soft reasoning over narrative worlds. The state variables are people, objects, locations, roles, observations and contradictions. Useful dialects therefore look less like algebra and more like a compact ledger of candidates and constraints. Unlike hard math, the fragile state is often a narrative constraint rather than a symbolic equation. This is why contrastive state-consistency dialects help: they force the listener to reject options that violate any recorded observation before emitting the final choice.

A.14 Dialect-card field definitions

A dialect card is stored as text, but it has a structured internal contract. The symbol inventory defines compact markers and their intended operations. The grammar defines the order in which the listener should read state, derivation and answer fields. The operator list defines reusable transformations such as option elimination, equation binding, contradiction testing, evidence ranking, subgoal decomposition and verifier invocation. The usage rules define when the dialect should be used and when it should be avoided. The failure-mode field records cases where the dialect produced wrong answers, ambiguous parses or excessive compression. The profile field records empirical accuracy, generated-token statistics, listener compatibility, publicness and task tags.

This structure distinguishes a dialect card from a prompt string. A prompt string is usually evaluated by whether it makes one model perform better on one distribution. A dialect card is evaluated by whether it can be reused, transferred, resisted, composed and archived. The same card can be a private self-dialect for one model, a public teaching dialect for another, and a risky imported dialect for a third. This relational identity is the reason that MDia requires speaker–listener matrices and not only aggregate accuracy.

A.15 Benchmark-level dialect operators

The eight benchmarks select for different symbolic operators. The purpose of a dialect is therefore not to shorten every trace with the same abbreviation rule. It is to preserve the type of intermediate state that the listener needs for a particular family of queries. Table 25 summarizes the operators that appear in useful dialect cards and the risk associated with over-aggressive compression.

Table 25: Benchmark-level dialect operators. These operators are not manually fixed formal languages; they summarize recurring slots and constraints that useful LSF cards tend to preserve.

Benchmark	Useful dialect operators	What the operator preserves	Overcompression risk
MMLU-Pro	Option state; elimination mark; domain tag; confidence update; final-answer marker.	Compact comparison among plausible answer options.	Removing evidence markers can make two plausible options indistinguishable.
GPQA-main	Assumption tag; scientific relation; contradiction mark; evidence-rank operator; uncertainty flag.	Expert knowledge discrimination under adversarial distractors.	Short traces can hide the assumption that determines the correct answer.
MATH500	Variable binding; equation transform; subgoal label; algebraic checkpoint; answer-normalization tag.	Algebraic state and derivation consistency.	Dropping variable bindings can preserve a plausible path but break the final expression.
AIME	Case split; modular constraint; boundary test; verifier tag; integer-answer check.	Multi-step search, calculation discipline and final numerical validation.	Very terse traces can delete the verification step needed for hard problems.
BFCL-v3	Function-name binding; argument-slot alignment; default-value guard; parallel-call splitter.	The exact API schema and the number/order of tool calls.	A compressed answer can be semantically right but unparsable if one argument key or call boundary is lost.
LiveCodeBench-output	State trace; loop counter; branch marker; stdout normalizer; newline guard.	Program-execution state and exact output formatting.	Natural-language shortcuts can change whitespace, ordering or the final printed value.
MultiHopRAG	Entity bridge; evidence-id tag; support/contradiction mark; null-support guard.	Multi-document evidence sufficiency and the distinction between supported and unanswerable queries.	Compression can collapse two hops into an unsupported guess.
MuSR	Candidate ledger; state table; observation tag; contradiction test; role/allocation schema.	Narrative consistency over multiple entities and latent constraints.	Dropping one observation can leave a wrong option globally consistent with the remaining trace.

These operators also explain why the same dialect may transfer differently across task families. A public MMLU-Pro dialect can be short because option discrimination often needs a small state vector. A public AIME dialect may require more explicit verification because a single omitted constraint can change the final integer answer. For BFCL-v3 and LiveCodeBench-output, the final parser-facing format is part of the reasoning state; for MultiHopRAG and MuSR, the fragile state is evidence sufficiency or narrative consistency. The router therefore estimates whether the query needs option tracking, verification, schema binding, execution-state simulation, evidence guarding or narrative-state consistency before choosing a compact direct dialect, a bridge dialect or a planner-solver-verifier composition.

A.16 Representative LSF-card skeletons

The paper does not require a single canonical LSF syntax. Different speakers may invent different surface symbols, and MDia evaluates those dialects empirically. Nevertheless, the following skeletons clarify what is meant by a reusable dialect card. They are templates for the card fields, not hand-authored final dialects.

Table 26: Representative LSF-card skeletons. The skeletons illustrate the reusable fields that MDia asks a speaker to instantiate from high-leverage traces.

Card type	Symbol inventory	Grammar sketch	Routing cue
Evidence dialect	C_i for candidate, E^+ for support, E^- for contradiction, ! for decisive evidence.	List candidates, attach compact evidence tags, eliminate tensions, emit final option.	Useful for knowledge-discrimination tasks and open listeners.
Algebra dialect	v for variable, Δ for transform, $=$ for invariant, ✓ for verified step.	Bind variables, apply transformations, check invariant, normalize answer.	Useful for MATH500-like derivations.
Verifier dialect	*case* for branch, *mod* for modular constraint, *bd* for boundary, *chk* for final check.	Enumerate cases, test constraints, reject invalid branches, verify final answer.	Useful for AIME-like hard math and resistant listeners that need explicit checks.
Bridge dialect	Short public symbols plus redundant final-answer and evidence slots.	Translate a private dialect into an adoptable intermediate form before execution.	Useful when direct foreign-dialect risk is high.
Schema dialect	fn for function, arg_k for argument slot, $\|$ for parallel calls, def for defaults.	Bind schema, align arguments, split requested calls, emit only a parseable call list.	Useful for BFCL-v3-style tool-call generation.
Execution dialect	s_t for state, *br* for branch, i for loop index, *out* for stdout.	Simulate the program state, normalize printed output, emit only exact stdout.	Useful for LiveCodeBench-output, where the answer contract is exact text.
Evidence-with-guard dialect	h_1, h_2 for hops, *sup* for support, *null* for insufficiency.	Bridge entities across documents, verify both hops, return null when support is absent.	Useful for MultiHopRAG and retrieval-style QA.
Narrative-state dialect	p_i for person/object, loc_t for state, *obs* for observation, $\perp$ for contradiction.	Maintain a compact state ledger, eliminate impossible options, emit the consistent choice.	Useful for MuSR, where hidden narrative constraints dominate.

The skeletons are intentionally simple. The important point is that a dialect card contains a contract that can be stored, retrieved, scored and reused. A surface prompt without stable fields cannot support publicness, listener openness, teaching advantage or dialect survival analysis.

A.17 Profile construction details

The profile vector for model M_i combines direct reasoning features and dialect-transfer features. The direct features include raw accuracy and generated-token counts on the eight benchmark columns; hard-math and knowledge-strength summaries are then computed from the relevant subsets. The hard-math strength feature averages normalized MATH500 and AIME accuracy. The knowledge strength feature averages normalized MMLU-Pro and GPQA-main accuracy. Verbosity is the normalized mean generated-token count under Raw CoT.

The transfer features are computed from validation-style single-dialect matrices. Foreign openness is the average utility gain from imported dialects. Foreign resistance is the share of imported dialects that underperform the listener's raw or self-dialect baseline. Speaker publicness is the average utility of a speaker's dialect across foreign listeners. Teaching advantage is the difference between a speaker's foreign-listener utility and the target listener's self-dialect utility. Compression tolerance measures whether a listener preserves accuracy when generated tokens fall sharply. Audience specificity is the variance of a speaker's dialect utility across listeners.

The profile labels in Table 2 are descriptive summaries of this vector. They are intentionally not treated as fixed model identities. If a future model has the same hard-math strength, verbosity, openness and publicness pattern, MDia should make the same type of prediction for it. This is the reason that the Results section emphasizes compact expert reasoners, verbose expert reasoners, open listeners and public speakers rather than naming a model as the scientific finding.

A.18 Validation-test separation and provenance audit

The manuscript separates three evidence streams. The first stream is the held-out accuracy–token table, which supports the main performance claim. The second stream is the split-validation route-selection audit, which tests whether route selection can be fixed on a router-validation partition and evaluated on a held-out partition. The third stream is validation-style mechanism analysis, including speaker–listener matrices, model profiles, rule validation, and functional influence. The paper never averages a transfer-matrix entry with a router-based held-out result as if they were the same object.

This separation is essential because the empirical question differs across streams. The held-out table asks whether the final MDia workflow improves the accuracy–token frontier. The split-validation route-selection audit asks whether archived evidence can select routes without using the evaluation half. The transfer matrix asks whether a dialect generated by speaker j is useful for listener i. The rule bank asks whether a sociolinguistic hypothesis is compatible with the validation-style evidence. The functional influence proxy asks whether removing a source family changes the utility of the archived dialect pool. Conflating these streams would make the claims harder to audit. Keeping them separate makes the paper more conservative and more reproducible.

Every reported table value is backed by a source path, benchmark name, listener, method label, evaluator version, and saved-output count. Records with incomplete outputs or incompatible benchmark counts are excluded from the main claims. When a row is missing from the selected/profiled stream, the table marks it rather than filling it with a different protocol.

A.19 Interpretability and safety considerations

Machine dialects are designed for efficient model use, not for direct human explanation. This creates a trade-off. A compact dialect can reduce decoding cost and preserve state for the listener, but it may be less transparent to a human auditor than a natural-language rationale. A deployment system should, therefore, distinguish internal reasoning dialects from external explanations. The final answer can be accompanied by a human-readable summary generated after the machine-dialect computation has finished. The summary should be evaluated separately from the internal dialect trace.

The social-router design can also improve safety. If a listener is compression-fragile, the router can avoid ultra-compact dialects. If a task is high-stakes, the router can require a verifier dialect or fall back to Raw CoT. If a dialect family shows high foreign-dialect risk for a listener, the archive can lower its route score under Eq. 19. These safeguards follow directly from treating dialect use as a measured speaker–listener relation.

A.20 How MDia differs from prompt-compression methods

A pure prompt-compression work would ask how to make a single model produce shorter rationales. MDia asks how a heterogeneous society of models develops reusable dialects and how another model should decide which dialect to use. The difference appears in the data structures: MDia records speakers, listeners, dialect families, transfer profiles, publicness, openness, and route plans. It appears in the experiments: MDia includes speaker–listener matrices, split-validation route-selection audits, rule validation, and functional influence analysis. It appears in the claims: the paper does not assert that one compressed trace format is universally optimal. It argues that dialect utility is receiver-relative, task-conditioned and profile-dependent.

This distinction also explains why MDia can preserve strong main results. A dialect that harms one listener can still be public for another listener. A concise dialect that performs well on MMLU-Pro can be insufficient for AIME if it removes verification state. The archive turns these observations into routing evidence rather than hiding them inside an aggregate metric.

A.21 Representative machine-sociolinguistic phenomena

Validation-style analyzes identify several recurring dialect phenomena that are useful to interpret the framework. The first phenomenon is *public dialect asymmetry*. A public dialect is not simply a dialect with high self-use accuracy. It is a dialect whose conventions can be adopted by several listeners whose internal reasoning procedures differ from the speaker's procedure. This distinction matters because a low-cost or unstable speaker can still create a compact convention that is broadly legible to other models. The mechanism resembles an audience-facing register: the dialect sacrifices some idiosyncratic internal richness in exchange for a more portable notation. For MDia, this means that the router should not rank dialects by speaker strength alone. It should estimate how much a dialect travels across listeners.

The second phenomenon is *listener openness*. Some listeners reliably benefit from imported dialects, while others are more effective when they use a self-dialect or a dialect created by a closely matched profile. This is not captured by a single aggregate benchmark score. Two listeners with similar raw accuracy can differ in their response to the same foreign dialect because one listener treats the symbols as useful state compression, whereas the other treats them as constraints that interrupt its own reasoning path. Listener openness is therefore a property of the listener–dialect interface, not a property of the listener in isolation.

The third phenomenon is *teaching advantage*. A dialect can be a better teaching language than a self-talk language. This happens when the speaker's own internal procedure does not need the explicit markers that make the dialect useful to other listeners. For example, a speaker may introduce subgoal markers, verifier tags, or option-elimination symbols that are redundant for itself but useful for a listener that would otherwise produce a longer natural-language trace. The phenomenon supports the central machine-dialectology view: dialects are communicative artifacts whose value depends on the audience.

The fourth phenomenon is *specialist resistance*. A strong listener may reject a compact imported dialect when the dialect removes the intermediate state that the listener normally uses for hard reasoning. This pattern is especially important for hard-math tasks. A shorthand that works on option discrimination can be too thin for AIME-like derivations, unless it preserves constraint checks and verification operators. The router should therefore treat resistance as a routing signal. A resistant expert should either use a self-dialect, a bridge dialect, or a verifier-rich composition.

The fifth phenomenon is *controlled code-switching*. A single dialect is often sufficient for low-difficulty cases, but hard queries benefit from switching among roles. A planner dialect can decompose the problem, a solver dialect can perform compact transformations, and a verifier dialect can test the final answer. This is not an ensemble for its own sake. It is a structured route through dialect roles, selected because different symbolic conventions preserve different parts of the reasoning state.

These phenomena explain why the manuscript gives a large role to validation-style mechanism analysis. The main table answers the performance question. The speaker–listener matrices and rule bank answer the scientific question of why a route works for one model community and not another. Together, they make MDia a study of machine dialects rather than a collection of shortened prompts.

A.22 Display-item and table audit

All main-text display items serve one of three roles. Table 1 reports the primary held-out accuracy–token comparison. Tables 2, 3, 4, and 7 provide mechanism evidence for receiver-relative dialect utility. The supplementary tables and figures provide provenance, route-selection, transfer, rule-bank, and robustness analyses. This separation is intended to prevent mechanism-validation records from being interpreted as direct replacements for the held-out main-table evaluation.

A.23 How to read the full rule bank

The full rule bank is not a list of axioms. It is a falsifiable inventory of sociolinguistic hypotheses translated into machine-dialect predictions. A rule may be strongly supported in the current archive, weakly supported, or incompatible with the current metric. The purpose of reporting the complete bank is to make the scientific scope clear. MDia does not claim that every analogy from human sociolinguistics transfers cleanly to LLM communities. It claims that such analogies can be operationalized, measured and revised.

A useful rule has three components. First, it names a social mechanism, such as publicness, openness, accommodation, resistance, teaching advantage or code-switching. Second, it states a machine prediction in terms of measurable records: accuracy deltas, generated-token deltas, transfer asymmetry, route selection, profile dependence or survival of dialect families. Third, it specifies the evidence stream on which the prediction is tested. Main-table evidence, split-validation route-selection evidence and validation-style transfer evidence are not interchangeable.

The supported rules are used in the paper to explain routing choices. For example, public-dialect asymmetry supports selecting a dialect that travels well even when its speaker is not the strongest raw reasoner. Listener-openness asymmetry supports listener-specific routing rather than a universal best dialect. Foreign-dialect risk supports bridge or self-dialect fallback for resistant listeners. Cost-sensitive utility supports route scoring that includes generated tokens and not only correctness. Compression-fragility supports avoiding ultra-compact traces for listeners that lose necessary state under aggressive abbreviation.

The weaker rules are still useful because they prevent overgeneralization. A rule can fail when its human analogy is too broad, when the available metric is too coarse, or when the benchmark distribution stresses a different reasoning property. For example, a simple shortest-dialect principle is not sufficient for hard math because verification state can be more important than brevity. These exceptions show that MDia studies measured social behavior among model profiles.